

MATHEMATICS FOR COMMUNICATION ENGINEERING

(EC241AT)

Notes Prepared by

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Reference Textbooks

1. *Digital Communication Systems*, Simon Haykin, Wiley, Standard Edition, 2013
2. *Analog and Digital Communications*, Hwei P. Hsu, 2nd Edition, 2003.
3. Lecture Notes of Dr. S. Ravishankar, Professor, Department of ECE, RVCE.

Unit-5

Communication Through AWGN Channels

Detection and Estimation

Detection is a deterministic process and estimation is a random process.

Detection

– Converting a waveform to a test statistic.

Estimation

– Mapping test statistics to a symbol.

At the Receiver

$$r(t) = s_i(t) + n(t) \quad (1)$$

$s_i(t)$ is a deterministic signal with finite energy and finite bandwidth

$n(t)$ is a random signal with infinite bandwidth and infinite energy

$r(t)$ is a random signal with infinite bandwidth and infinite energy

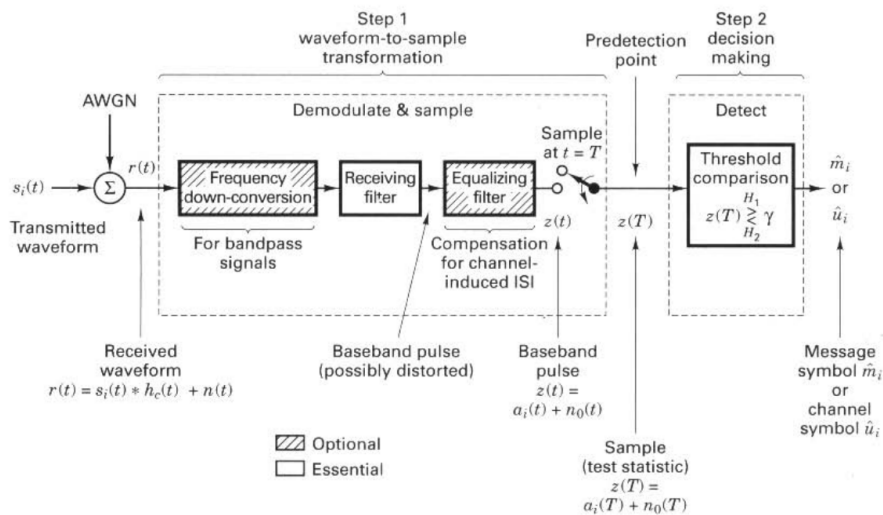


Figure 1: Two basic steps in demodulation/detection of digital signals

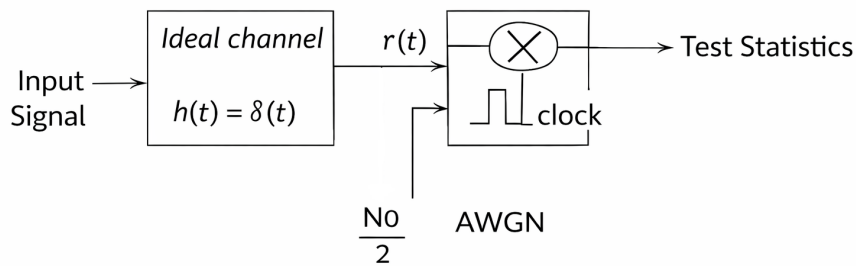
Detection

Detection is carried out using three types:

1. Clock sampling / Center sampling

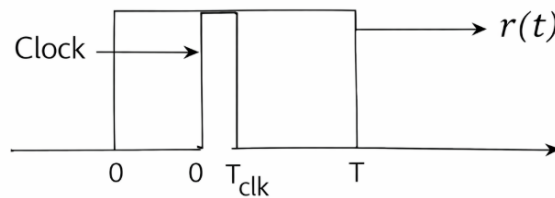
2. Averaging
3. Matched filter

Clock Sampling



At the receiver,

$$r(t) = s_i(t) + n(t) \quad (2)$$



Clock signal:

Clock: rectangular pulses of width T_{clk}

$$T_{\text{clk}} \ll T \quad (\text{symbol duration})$$

$\therefore$ Clock bandwidth $\gg$ Signal bandwidth

In clock sampling, the noise outside the signal range also contributes to the sampled value. Sampling is performed only at the center of the pulse.

Test statistics is noisy in clock sampling.

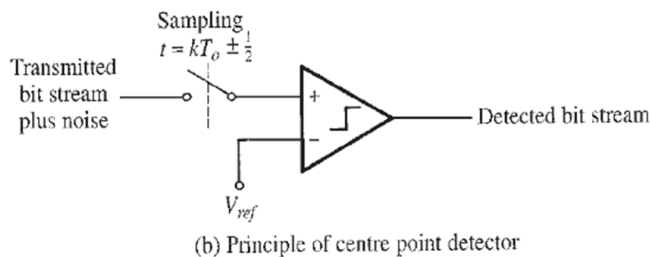
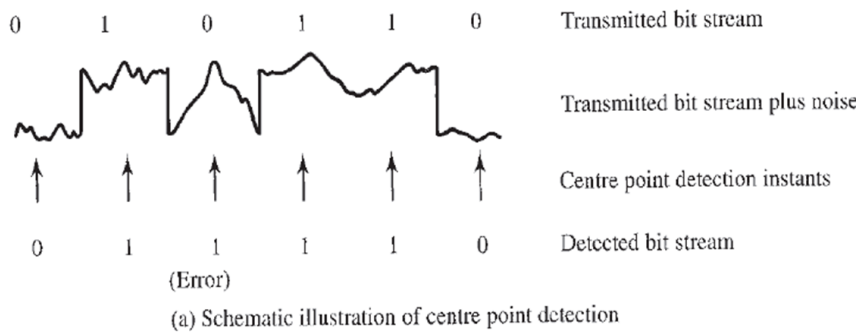


Figure 2: Clock Sampling

Averaging

Sample the signal at multiple instants from 0 to T and then average it. Noise also gets averaged.

$$r(t) = s_i(t) + n(t) \quad (3)$$

$$\text{Mean of } n(t) = 0$$

$$\text{Mean of } r(t) = \text{Mean of } s_i(t)$$

Test statistics is cleaner in averaging as compared to clock sampling. In averaging, we cannot extract the full energy of $r(t)$.

After averaging, the signal is

$$r'_i = \frac{1}{N} \sum_{i=1}^N r_i \quad (4)$$

When SNR is high and the communication distance is short (e.g., RS-232, 15 m cable), clock sampling or averaging is used for detection.

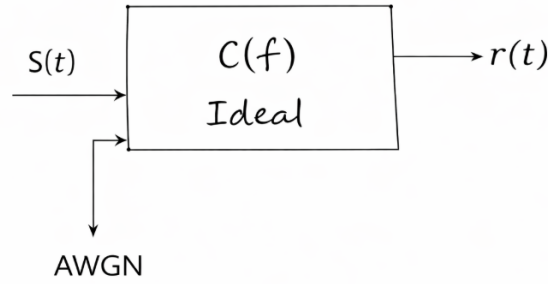
Block representation: with

$$|C(f)| = 1$$

$$R(f) = S(f)C(f) \quad (5)$$

$$r(t) = \Re \{ \mathcal{F}^{-1} [R(f)] \} = \Re \{ \mathcal{F}^{-1} [|S(f)| |C(f)| e^{j(\angle S(f) + \phi)}] \} \quad (6)$$

$$r(t) = \Re \{ \mathcal{F}^{-1} [|S(f)| e^{j(\angle S(f) + \phi)}] \} \quad (7)$$



If we can match

$$\angle \phi = \angle S(f), \quad (8)$$

then we can extract the full energy.

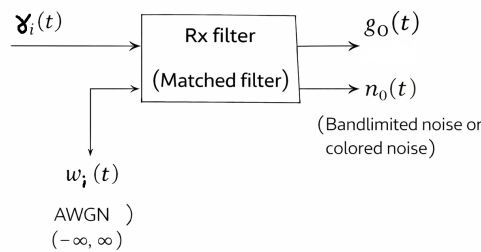
When SNR is low and in long distance communication, we use a matched filter.

Matched Filter

This restricts the noise to the signal bandwidth. It provides a conjugate for the signal so that we get maximum SNR at the output of the matched filter.

In a matched filter, the test statistic is cleaner, SNR improves, and there will be a better estimation.

Block representation:



Noise input:

$$w_i(t) \quad (\text{AWGN})$$

Output noise:

$$n_o(t) \quad (\text{bandlimited or colored noise})$$

Received signal:

$$r_i(t) = g_i(t) + w_i(t) \quad (9)$$

Output SNR:

$$\left(\frac{S}{N} \right)_{i/p} = \frac{|g_i(T)|^2}{E[w_i^2(t)]} \quad (10)$$

$$\left(\frac{S}{N} \right)_{o/p} = \frac{|g_o(T)|^2}{E[n_o^2(t)]} \quad (11)$$

At input:

Noise level is more than the signal level.

At output:

Signal level becomes higher than noise level.

Need to establish two conditions:

1. What is the maximum SNR possible from a matched filter?
2. What should be the shape of the matched filter to achieve maximum SNR at the output?

Frequency Domain SNR

The output of the matched filter is

$$g_o(t) = g_i(t) * h(t) \quad (12)$$

$$G_o(f) = G_i(f) H(f) \quad (13)$$

$$g_o(t) = \int_{-\infty}^{\infty} H(f) G_i(f) e^{j2\pi ft} df \quad (14)$$

The matched filter is sampled at $t = T$.

Signal energy at the output:

$$|g_o(T)|^2 = \left| \int_{-\infty}^{\infty} H(f) G_i(f) e^{j2\pi fT} df \right|^2 \quad (15)$$

PSD of noise at the output:

$$S_N(f) = \frac{N_0}{2} |H(f)|^2 \quad (16)$$

Output SNR:

$$\left(\frac{S}{N} \right)_{o/p} = \frac{\left| \int_{-\infty}^{\infty} H(f) G_i(f) e^{j2\pi fT} df \right|^2}{\int_{-\infty}^{\infty} S_N(f) df} \quad (17)$$

Apply Cauchy-Schwarz inequality: If $\phi_1(x)$ and $\phi_2(x)$ are two complex functions, then the integral of their product is less than or equal to the product of their individual integrals. Equality holds when ϕ_1 and ϕ_2 are conjugates of each other.

$$\int_{-\infty}^{\infty} |\phi_1(x)|^2 dx < \infty, \quad \int_{-\infty}^{\infty} |\phi_2(x)|^2 dx < \infty \quad (18)$$

then

$$\left| \int_{-\infty}^{\infty} \phi_1(x) \phi_2(x) dx \right|^2 \leq \left(\int_{-\infty}^{\infty} |\phi_1(x)|^2 dx \right) \left(\int_{-\infty}^{\infty} |\phi_2(x)|^2 dx \right) \quad (19)$$

Equality holds when

$$\phi_1(x) = C \phi_2^*(x) \quad (20)$$

Let

$$\phi_1(x) = H(f), \quad \phi_2(x) = G_i(f) e^{j2\pi fT} \quad (21)$$

—

Properties of a Matched Filter

1. The spectrum of the MF output is proportional to the energy spectral density of the input signal:

$$S_o(f) = H_{\text{opt}}(f)S(f) \quad (22)$$

$$S_o(f) = |S(f)|^2 e^{-j2\pi fT} \quad (23)$$

2. The MF output is a time-reversed autocorrelation function of the input:

$$\mathcal{F}^{-1} \{s_o(t)\} = \mathcal{F}^{-1} \{|S(f)|^2 e^{-j2\pi fT}\} \quad (24)$$

$$s_o(t) = R_s(T - t) \quad (25)$$

3. Output SNR of matched filter:

$$\left(\frac{S}{N}\right)_{\text{o/p}} = \frac{2E}{N_0} \quad (26)$$

SNR depends only on signal energy and not on waveform shape.

4. Matching conditions:

$$|H(f)| = |S(f)| \quad (27)$$

$$\angle H_{\text{opt}}(f) = -\theta(f) - 2\pi fT \quad (28)$$

Correlator Realization of a Matched Filter

Limitations of matched filter:

1. For M -symbols, M matched filters are required.
2. MF operation is a convolution.
3. MF output is continuous from 0 to $2T$.

Output of matched filter:

$$z(t) = r(t) * h(t) \quad (29)$$

$$r(t) = s(t) + \omega(t) \quad (30)$$

$$z(t) = \int_0^T r(\tau)h(t - \tau) d\tau \quad (31)$$

Matched filter impulse response:

$$h(t) = k s(T - t) \quad (32)$$

Substituting:

$$z(t) = \int_0^T r(\tau)s(T - (t - \tau)) d\tau \quad (33)$$

At $t = T$:

$$z(T) = \int_0^T r(\tau)s(\tau) d\tau \quad (34)$$

Thus the matched filter is equivalent to a correlator.

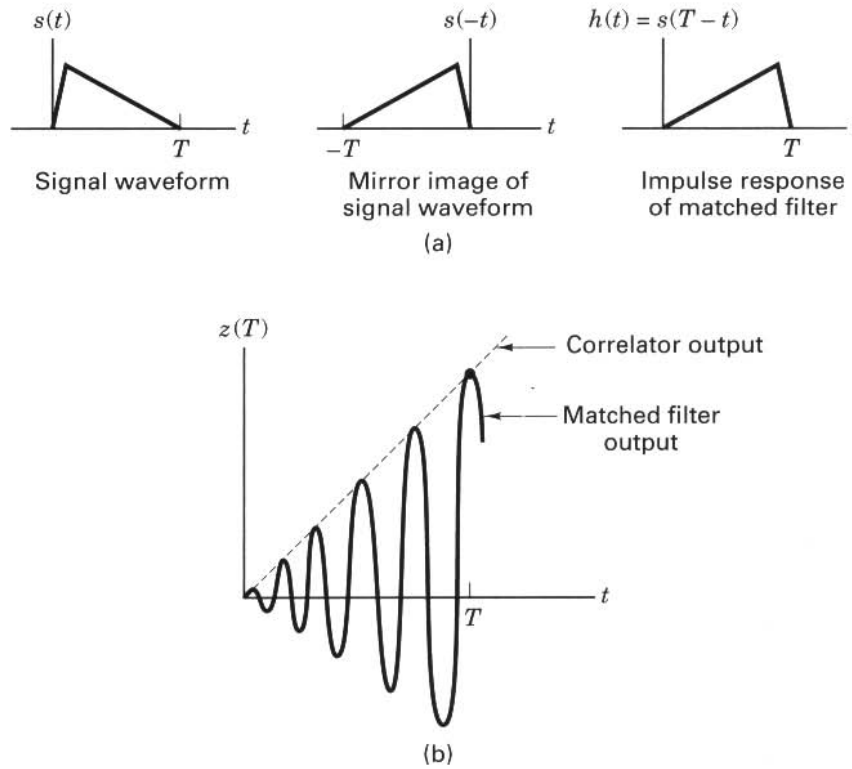


Figure 3: Correlator and matched filter. (a) Matched filter characteristic. (b) Comparison of correlator and matched filter outputs.

Estimation

Estimation is the process of mapping test statistics to a symbol.

Receiver Model

Sequence of bits $\{m_1, m_2, \dots, m_k\} \rightarrow$ Symbols $\{s_1, s_2, \dots, s_M\}$

$$r(t) = s_i(t) + w(t) \quad (35)$$

where $w(t)$ is AWGN.

At the detector, the test statistic is

$$z(T) = s_i(T) + w(T) \quad (36)$$

(signal sample + noise sample)

Estimated symbol:

$$\hat{s} = s_i, \quad i = 1, 2, \dots, M$$

(MAP or ML decision)

Statistical Events

Two statistical events are observed:

1. A priori probability: $p(s_i)$ at the transmitter
2. A posteriori probability: $p(s_i|z)$

Using Bayes' theorem:

$$p(s_i|z) = \frac{p(z|s_i)p(s_i)}{p(z)} \quad (37)$$

where

$$p(z) = \sum_i p(z|s_i)p(s_i) \quad (38)$$

is a common factor and can be dropped for decision purposes.

Thus,

$$p(s_i|z) \propto p(s_i)p(z|s_i) \quad (39)$$

For equiprobable symbols:

$$p(s_i) = \frac{1}{M} \quad (40)$$

Maximum likelihood (MLI) decision:

$$\text{MLI: } p(z|s_i) \text{ is maximized}$$

Maximum a posteriori (MAP) decision:

$$\text{MAP: } p(s_i|z) \text{ is maximized}$$

Either maximizing MLI or MAP will give the minimum probability of error.

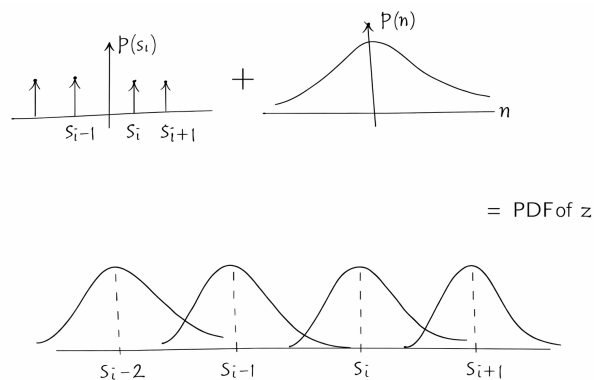
In practice, MLI is easier to compute.

Likelihood Probability $p(z|s_i)$

$$z = s_i + n \quad (41)$$

- $s_i \rightarrow$ uniform PDF
- $n \rightarrow$ Gaussian PDF

Thus, z has a Gaussian PDF centered at s_i .



Mean and Variance

$$\text{Mean}[z] = E[s_i + n] = E[s_i] \quad (\text{signal mean}) \quad (42)$$

$$\text{Var}[z] = E[(z - \bar{z})^2] = E[(s_i + n - s_i)^2] = E[n^2] = \sigma^2 = \frac{N_0}{2} \quad (43)$$

Hence,

$$\text{Var}[z] = \text{variance of noise.}$$

Error Probability for Symbol s_i

$$P_{e,s_i} = p(s_i \text{ sent and } z \neq s_i) \quad (44)$$

$$P_{e,s_i} = p(s_i) p(z \neq s_i | s_i) \quad (45)$$

(Two independent events)

$$P_{e,s_i} = p(s_i) [1 - p(z = s_i | s_i)] \quad (46)$$

But from Bayes' theorem:

$$p(z = s_i | s_i) p(s_i) = p(s_i | z) p(z) \quad (47)$$

Therefore,

$$P_{e,s_i} = p(s_i) - p(z) p(s_i | z) \quad (48)$$

which shows the relation between prior and posterior probabilities. **Total Probability of Error**

$$P_e = \sum_{i=1}^M P_{e,s_i} = \sum_{i=1}^M p(s_i) p(z \neq s_i | s_i) \quad (49)$$

$$= \sum_{i=1}^M p(s_i) [1 - p(z = s_i | s_i)] \quad (50)$$

$$= \sum_{i=1}^M p(s_i) - \sum_{i=1}^M p(s_i) p(z = s_i | s_i) \quad (51)$$

Since

$$\sum_{i=1}^M p(s_i) = 1, \quad (52)$$

we obtain

$$P_e = 1 - \sum_{i=1}^M p(s_i) p(z = s_i | s_i) \quad (53)$$

The term

$$p(z = s_i | s_i) \quad (54)$$

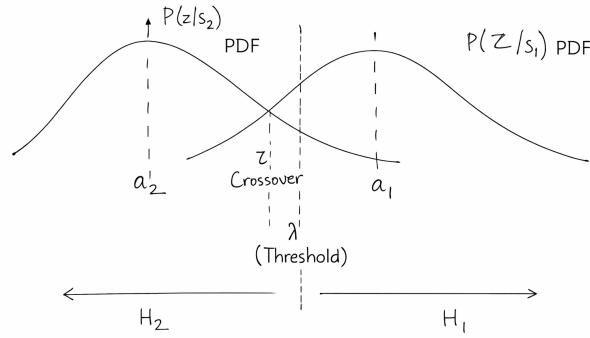
represents the CDF over all possible $z \in s_i$.

Thus,

$$P_e = 1 - \sum_{i=1}^M p(s_i) \int_{z=s_i} p(z|s_i) dz \quad (55)$$

Therefore, using either MLI or MAP decision rules will reduce (minimize) the probability of error.

Probability of Error for Two-Symbol Case



Let the decision threshold be λ .

$$P_e = p(s_1) P(H_2|s_1) + p(s_2) P(H_1|s_2) \quad (56)$$

$$= p(s_1) \int_{-\infty}^{\lambda} p(z|s_1) dz + p(s_2) \int_{\lambda}^{\infty} p(z|s_2) dz \quad (57)$$

$$= p(s_1) \int_{-\infty}^{\lambda} p(z|s_1) dz + p(s_2) \left[1 - \int_{-\infty}^{\lambda} p(z|s_2) dz \right] \quad (58)$$

$$= p(s_2) + \int_{-\infty}^{\lambda} [p(s_1)p(z|s_1) - p(s_2)p(z|s_2)] dz \quad (59)$$

To find the threshold λ that minimizes the probability of error:

$$\frac{dP_e}{d\lambda} = 0 \quad (60)$$

Let

$$f(z, u) = p(s_1)p(z|s_1) - p(s_2)p(z|s_2) \quad (61)$$

Using Leibniz's rule:

$$\frac{d}{du} \int_{a(u)}^{b(u)} f(z, u) dz = f(b(u), u) \frac{db(u)}{du} - f(a(u), u) \frac{da(u)}{du} + \int_{a(u)}^{b(u)} \frac{\partial f(z, u)}{\partial u} dz = 0 \quad (62)$$

Here,

$$a(\lambda) = -\infty, \quad b(\lambda) = \lambda \quad (63)$$

$$\frac{d}{du} \int_{-\infty}^{\lambda} f(z, u) dz = f(\lambda, u) \frac{d\lambda}{du} - f(-\infty, u) \frac{d(-\infty)}{du} + \int_{-\infty}^{\lambda} \frac{\partial f(z, u)}{\partial u} dz = 0 \quad (64)$$

Since $f(z, u)$ is independent of λ , the second and third terms vanish:

$$f(\lambda, u) \frac{\delta\lambda}{\delta u} = 0 \quad (65)$$

$$p(s_1) p(z = \lambda_0|s_1) - p(s_2) p(z = \lambda_0|s_2) = 0 \quad (66)$$

$$p(z = \lambda_0 | s_1) = \frac{p(s_2)}{p(s_1)} p(z = \lambda_0 | s_2) \quad (67)$$

Assuming Gaussian PDFs:

$$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(\lambda_0 - a_1)^2}{2\sigma^2}} = \frac{p(s_2)}{p(s_1)} \quad (68)$$

$$e^{-\frac{(\lambda_0 - a_1)^2 - (\lambda_0 - a_2)^2}{2\sigma^2}} = \frac{p(s_2)}{p(s_1)} \quad (69)$$

$$e^{\left[\frac{\lambda_0(a_1 - a_2)}{\sigma^2} - \frac{a_1^2 - a_2^2}{2\sigma^2} \right]} = \frac{p(s_2)}{p(s_1)} \quad (70)$$

Taking natural logarithm:

$$\frac{\lambda_0(a_1 - a_2)}{\sigma^2} - \frac{a_1^2 - a_2^2}{2\sigma^2} = \ln\left(\frac{p(s_2)}{p(s_1)}\right) \quad (71)$$

Solving for λ_0 :

$$\lambda_0 = \frac{a_1 + a_2}{2} + \frac{\sigma^2}{a_1 - a_2} \ln\left(\frac{p(s_2)}{p(s_1)}\right) \quad (72)$$

This is the threshold for minimum probability of error.

If

$$p(s_1) = p(s_2) = \frac{1}{2}, \quad (73)$$

then

$$\lambda_0 = \frac{a_1 + a_2}{2}. \quad (74)$$

From Eq. (72),

$$\lambda = \frac{v_1 + v_2}{2} + \frac{\sigma^2}{v_1 - v_2} \ln\left(\frac{p(s_2)}{p(s_1)}\right) \quad (75)$$

$$= \tau + \frac{\sigma^2}{v_1 - v_2} \ln\left(\frac{p(s_2)}{p(s_1)}\right) \quad (76)$$

From Eq. (57),

$$P_e = p(s_1) \int_{-\infty}^{\lambda} \frac{1}{\sqrt{2\pi\sigma}} \exp\left[-\frac{(z - v_1)^2}{2\sigma^2}\right] dz + p(s_2) \int_{\lambda}^{\infty} \frac{1}{\sqrt{2\pi\sigma}} \exp\left[-\frac{(z - v_2)^2}{2\sigma^2}\right] dz \quad (77)$$

Let

$$u = \frac{z - v_1}{\sqrt{2\sigma}}, \quad v = \frac{z - v_2}{\sqrt{2\sigma}} \quad (78)$$

$$dz = \sqrt{2}\sigma du, \quad dz = \sqrt{2}\sigma dv \quad (79)$$

Limits:

$$z \rightarrow -\infty \Rightarrow u \rightarrow -\infty, \quad z \rightarrow \infty \Rightarrow v \rightarrow \infty$$

$$z = \lambda \Rightarrow u = \frac{\lambda - v_1}{\sqrt{2}\sigma}, \quad z = \lambda \Rightarrow v = \frac{\lambda - v_2}{\sqrt{2}\sigma} \quad (80)$$

Thus,

$$P_e = p(s_1) \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\frac{-\lambda - v_1}{\sqrt{2}\sigma}} e^{-u^2} du + p(s_2) \frac{1}{\sqrt{\pi}} \int_{\frac{\lambda - v_2}{\sqrt{2}\sigma}}^{\infty} e^{-v^2} dv \quad (81)$$

Since the Gaussian function is symmetric,

$$P_e = \frac{p(s_1)}{2} \left[\frac{2}{\sqrt{\pi}} \int_{\frac{v_1 - \lambda}{\sqrt{2}\sigma}}^{\infty} e^{-u^2} du \right] + \frac{p(s_2)}{2} \left[\frac{2}{\sqrt{\pi}} \int_{\frac{\lambda - v_2}{\sqrt{2}\sigma}}^{\infty} e^{-v^2} dv \right] \quad (82)$$

$$P_e = \frac{1}{2} p(s_1) \operatorname{erfc} \left(\frac{v_1 - \lambda}{\sqrt{2}\sigma} \right) + \frac{1}{2} p(s_2) \operatorname{erfc} \left(\frac{\lambda - v_2}{\sqrt{2}\sigma} \right) \quad (83)$$

where λ is given by Eq. (76).

Case 1: When

$$p(s_1) = p(s_2), \quad (84)$$

then

$$\lambda = \tau = \frac{v_1 + v_2}{2}. \quad (85)$$

$$P_e = \frac{1}{4} \operatorname{erfc} \left(\frac{v_1 - 0.5(v_1 + v_2)}{\sqrt{2}\sigma} \right) + \frac{1}{4} \operatorname{erfc} \left(\frac{0.5(v_1 + v_2) - v_2}{\sqrt{2}\sigma} \right) \quad (86)$$

$$= \frac{1}{4} \operatorname{erfc} \left(\frac{v_1 - v_2}{2\sqrt{2}\sigma} \right) + \frac{1}{4} \operatorname{erfc} \left(\frac{v_1 - v_2}{2\sqrt{2}\sigma} \right) \quad (87)$$

$$P_e = \frac{1}{2} \operatorname{erfc} \left(\frac{v_1 - v_2}{2\sqrt{2}\sigma} \right) = \frac{1}{2} \operatorname{erfc} \left(\frac{\Delta v}{2\sqrt{2}\sigma} \right) \quad (88)$$

As $\Delta v \uparrow$, $\operatorname{erfc} \downarrow$, $P_e \downarrow$, $Q_0 S \uparrow$

Case 2: Unequal a priori probabilities

$$p(s_1) = p, \quad p(s_2) = 1 - p \quad (89)$$

$$\lambda = \tau + \frac{\sigma^2}{v_1 - v_2} \ln \left(\frac{1 - p}{p} \right) \quad (90)$$

$$P_e = \frac{p}{2} \operatorname{erfc} \left(\frac{v_1 - \lambda}{\sqrt{2}\sigma} \right) + \frac{1 - p}{2} \operatorname{erfc} \left(\frac{\lambda - v_2}{\sqrt{2}\sigma} \right) \quad (91)$$

Probability of error in terms of $d_{\min}$

As long as the distance between the symbols is maintained the same, the probability of error is invariant to rotation and translation of the coordinate system.

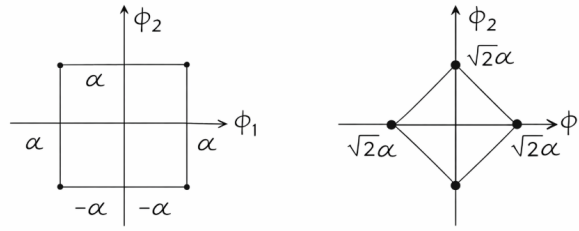


Figure 4: Rotation of the constellation

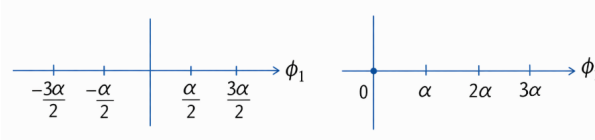
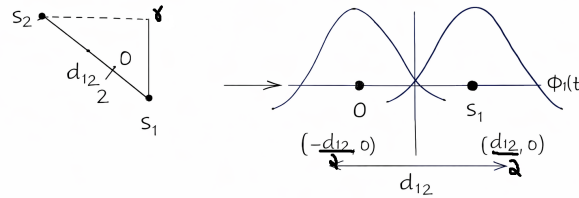


Figure 5: Translation of the constellation



$$P_e = \frac{1}{2} \left[\frac{2}{\sqrt{\pi}} \int_{\frac{d_{12}}{2\sqrt{2}\sigma}}^{\infty} e^{-v^2} dv \right] = \frac{1}{2} \operatorname{erfc} \left(\frac{d_{12}}{2\sqrt{2}\sigma} \right) \quad (92)$$

Since the minimum distance between symbols is $d_{\min}$,

$$P_e = \frac{1}{2} \operatorname{erfc} \left(\frac{d_{\min}}{2\sqrt{2}\sigma} \right) \quad (93)$$

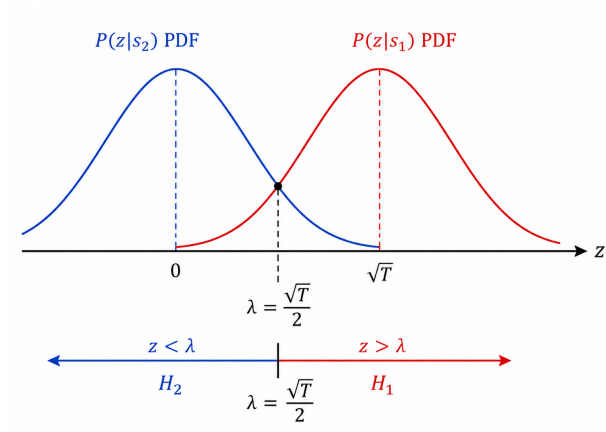
Probability of Error for Baseband Signals

$$P_e = \frac{1}{2} \operatorname{erfc} \left(\frac{d_{\min}}{2\sqrt{N_0}} \right), \quad \operatorname{erfc}(x) = \frac{e^{-x^2}}{\sqrt{\pi}} \quad (94)$$

Approximate equation:

$$P_e \approx \frac{1}{2\sqrt{\pi}} \exp \left(-\frac{d_{\min}^2}{4N_0} \right) \quad (95)$$

1. Unipolar Signal



$$d_{\min} = A\sqrt{T} \quad (96)$$

$$P_e = \frac{1}{2} \operatorname{erfc} \left(\frac{A\sqrt{T}}{2\sqrt{N_0}} \right) = \frac{1}{2} \operatorname{erfc} \left(\sqrt{\frac{E_{sav}}{2N_0}} \right) \quad (97)$$

$$E_{sav} = \frac{E_{s1} + E_{s2}}{2} = \frac{A^2 T}{2} \quad (98)$$

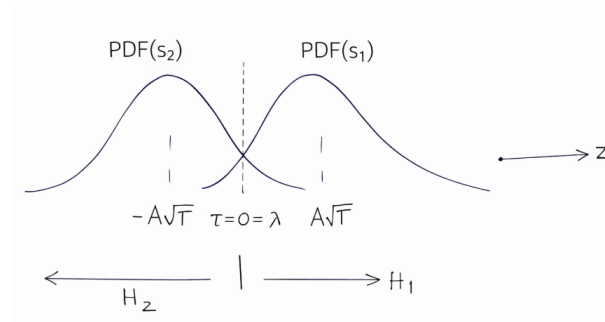
$$\text{As } \frac{E_{sav}}{2N_0} \uparrow \Rightarrow P_e \downarrow, Q_0 S \uparrow$$

Conversion formulas:

$$\operatorname{erfc}(x) = 2Q(\sqrt{2}x), \quad Q(x) = \frac{1}{2} \operatorname{erfc} \left(\frac{x}{\sqrt{2}} \right) \quad (99)$$

$$\operatorname{erfc}(x) \approx \frac{e^{-x^2}}{\sqrt{\pi}}, \quad Q(x) \approx \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \quad (100)$$

2. Polar Signal



$$d_{\min} = 2A\sqrt{T} \quad (101)$$

$$P_e = \frac{1}{2} \operatorname{erfc} \left(\frac{d_{\min}}{2\sqrt{N_0}} \right) = \frac{1}{2} \operatorname{erfc} \left(\frac{A\sqrt{T}}{\sqrt{N_0}} \right) = \frac{1}{2} \operatorname{erfc} \left(\sqrt{\frac{E_{sav}}{N_0}} \right) \quad (102)$$

$$E_{sav} = A^2T \quad (103)$$

$$\frac{E_{sav}}{N_0} \uparrow \Rightarrow \text{erfc} \downarrow \Rightarrow P_e \downarrow \Rightarrow \text{QoS} \uparrow$$

$$P_{e,\text{polar}} < P_{e,\text{unipolar}}$$

Polar signaling is more power efficient than unipolar signaling. Approximately 3 dB more energy is required in unipolar signaling to achieve the same error probability as polar signaling.

3. Manchester Signal

The probability of error for Manchester signaling is the same as that for the polar signal. The two symbols are antipodal in both polar and Manchester signaling.

$$P_e = \frac{1}{2} \text{erfc} \left(\frac{A\sqrt{T}}{\sqrt{N_0}} \right) = \frac{1}{2} \text{erfc} \left(\sqrt{\frac{E_{sav}}{N_0}} \right) \quad (104)$$

4. Bipolar Signal

$$P_e = (M-2) 2P_e P(0) + 2P_e P(+1) \quad (105)$$

where - $(M-2)$ inner symbols have errors on both sides, - the two outer symbols have errors in only one direction.

For $M = 3$:

$$P_e = 1 \cdot 2P_e \cdot \frac{1}{2} + 2P_e \cdot \frac{1}{4} = P_e + \frac{1}{2}P_e = \frac{3}{2}P_e \quad (106)$$

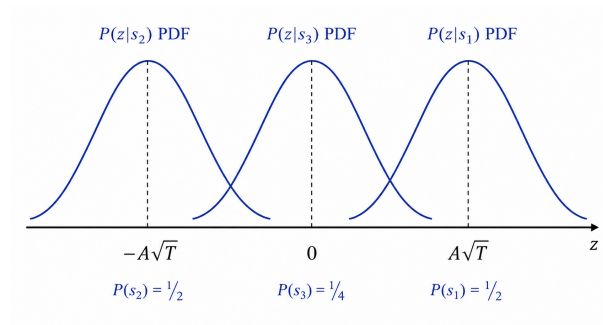
Thus,

$$P_e = \frac{1}{2} \cdot \frac{3}{2} \text{erfc} \left(\sqrt{\frac{E_{sav}}{2N_0}} \right) = \frac{3}{4} \text{erfc} \left(\sqrt{\frac{E_{sav}}{2N_0}} \right) \quad (107)$$

Since

$$E_{sav} = \frac{2A^2T}{3}, \quad (108)$$

$$P_e = \frac{3}{4} \text{erfc} \left(\sqrt{\frac{2A^2T}{3 \cdot 2N_0}} \right) = \frac{3}{4} \text{erfc} \left(\sqrt{\frac{A^2T}{3N_0}} \right) \quad (109)$$



Optimum Receiver Architecture

Choice of Symbols

Choice of symbols

$$\{s_1, s_2, \dots, s_M\} = \{s_m\}_{m=1}^M$$

based on

1. User rate
2. Medium characteristics

$\{s_m\}$ are functions of amplitude, frequency and phase

$$s_m = f(A, F, \phi)$$

Examples:

- Wireline
- Wireless
- Underwater

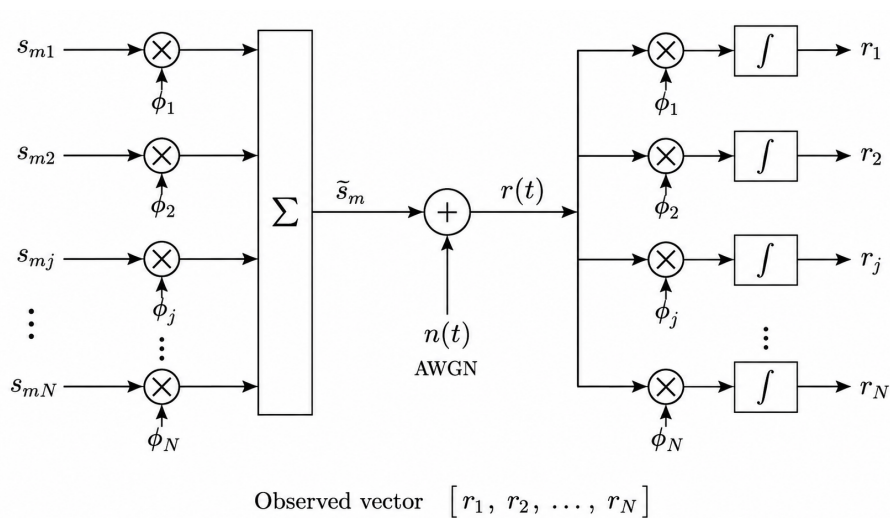
$$\{s_m\}_{m \in [1, M]}$$

can be represented as vectors in Euclidean space.

From $\{s_m\}$, we extract a basis of orthonormal functions

$$\{\phi_1, \phi_2, \dots, \phi_N\}, \quad N \leq M$$

If $N = M$, then $\{s_m\}$ are orthogonal.



Procedure for extraction is Gram-Schmidt orthogonalization.

$$s_m = \sum_{j=1}^N s_{mj} \phi_j$$

Tx and Rx keep a copy of the S -matrix.

No noise addition means S -matrix can be applied to received values of MF outputs.

$$\begin{bmatrix} s_{11} & \cdots & s_{1N} \\ \vdots & \ddots & \vdots \\ s_{M1} & \cdots & s_{MN} \end{bmatrix} \begin{bmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_N \end{bmatrix} = \begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_M \end{bmatrix}$$

Detection in Signal Space

$$r_j = s_{mj} + n_j$$

where n_j is a random variable.

But

$$\mathbf{r} = \mathbf{s}_m + \mathbf{n}(t)$$

or equivalently,

$$\mathbf{r} = \mathbf{s}_m + \bar{n}_1(t) + \bar{n}_2(t)$$

where $\bar{n}_2(t)$ is the noise component outside the Euclidean space.

The components of the observed vector $\mathbf{r}$ are

$$r_j = \langle \mathbf{r}, \phi_j \rangle$$

Since $r = s_m + n$,

$$r_j = \langle s_m, \phi_j \rangle + \langle n, \phi_j \rangle$$

Hence,

$$r(t) = \sum_{j=1}^N (s_{mj} + n_j) \phi_j + \bar{n}_2(t)$$

where

$$r_j = s_{mj} + n_j$$

To determine which s_m was transmitted, apply MAP detection.

Evaluate the a posteriori probability

$$P(s_m | \mathbf{r})$$

and choose the maximum.

Using Bayes rule,

$$P(s_m | \mathbf{r}) = \frac{P(\mathbf{r} | s_m) P(s_m)}{P(\mathbf{r})}$$

For equally likely symbols,

$$P(s_m) = \frac{1}{M}$$

and $P(\mathbf{r})$ is common for all cases. Hence,

$$\max_m P(s_m|\mathbf{r}) \equiv \max_m P(\mathbf{r}|s_m)$$

Thus, comparison of the likelihood functions $P(\mathbf{r}|s_m)$ is sufficient.

The conditional probability $P(\mathbf{r}|s_m)$ is a vector distribution with N components $(r_1, r_2, \dots, r_N)$.

Therefore, the following steps are required:

1. Find the pdf of $r_1, r_2, \dots, r_N$.
2. Check covariance $\text{Cov}(r_j, r_k)$ for statistical independence.
3. Check covariance $\text{Cov}(r_j, n_2)$.
4. Obtain the cost/decision rule.

1. Mean and Variance of r_j

$$E[r_j] = E[s_{mj} + n_j] = E[s_{mj}] + E[n_j]$$

Since s_{mj} is deterministic,

$$E[r_j] = s_{mj} + E[n_j]$$

Now,

$$n_j = \int_0^T n(t)\phi_j(t) dt$$

Hence,

$$E[n_j] = E\left[\int_0^T n(t)\phi_j(t) dt\right] = \int_0^T E[n(t)]\phi_j(t) dt$$

For zero mean noise,

$$E[n(t)] = 0$$

therefore,

$$E[n_j] = 0$$

and

$$E[r_j] = s_{mj}$$

Variance:

$$\text{Var}(r_j) = E[(r_j - s_{mj})^2] = E[n_j^2]$$

Now,

$$n_j = \int_0^T n(t)\phi_j(t) dt$$

thus,

$$\begin{aligned} E[n_j^2] &= E \left[\int_0^T n(t) \phi_j(t) dt \int_0^T n(\tau) \phi_j(\tau) d\tau \right] \\ &= \int_0^T \int_0^T \phi_j(t) \phi_j(\tau) E[n(t)n(\tau)] dt d\tau \end{aligned}$$

For AWGN,

$$E[n(t)n(\tau)] = \frac{N_0}{2} \delta(t - \tau)$$

Hence,

$$\text{Var}(r_j) = \frac{N_0}{2}$$

2. Covariance Between r_j and r_k

For $j \neq k$,

$$\text{Cov}(r_j, r_k) = E[r_j r_k] - E[r_j] E[r_k]$$

Substituting,

$$\begin{aligned} &= E[(s_{mj} + n_j)(s_{mk} + n_k)] - s_{mj} s_{mk} \\ &= E[s_{mj} s_{mk} + s_{mj} n_k + s_{mk} n_j + n_j n_k] - s_{mj} s_{mk} \end{aligned}$$

Since

$$E[n_j] = E[n_k] = 0$$

we obtain

$$\text{Cov}(r_j, r_k) = E[n_j n_k]$$

Now,

$$E[n_j n_k] = \int_0^T \int_0^T \phi_j(t) \phi_k(\tau) E[n(t)n(\tau)] dt d\tau$$

Using

$$E[n(t)n(\tau)] = \frac{N_0}{2} \delta(t - \tau)$$

gives

$$E[n_j n_k] = \frac{N_0}{2} \int_0^T \phi_j(t) \phi_k(t) dt$$

Since the basis functions are orthonormal,

$$\int_0^T \phi_j(t) \phi_k(t) dt = 0 \quad j \neq k$$

Therefore,

$$\text{Cov}(r_j, r_k) = 0$$

Hence, r_j and r_k are statistically independent.

3. Covariance Between r_j and n_2

$$\text{Cov}(r_j, n_2) = E[r_j n_2] - E[r_j]E[n_2]$$

Since $E[n_2] = 0$,

$$\text{Cov}(r_j, n_2) = E[r_j n_2]$$

Substituting $r_j = s_{mj} + n_j$,

$$= E[(s_{mj} + n_j)n_2]$$

$$= s_{mj}E[n_2] + E[n_j n_2]$$

Since $E[n_2] = 0$,

$$\text{Cov}(r_j, n_2) = E[n_j n_2]$$

Now,

$$n_j = \int_0^T n(t)\phi_j(t) dt$$

Hence,

$$E[n_j n_2] = E \left[n_2 \int_0^T n(t)\phi_j(t) dt \right]$$

Using orthogonality between the signal-space component and the noise component outside Euclidean space,

$$E[n_j n_2] = 0$$

Therefore,

$$\text{Cov}(r_j, n_2) = 0$$

Hence, the noise outside Euclidean space does not affect decisions.

Joint PDF

Since $r_1, r_2, \dots, r_N$ are statistically independent,

$$P(\mathbf{r}|s_m) = \prod_{k=1}^N P(r_k|s_{mk})$$

Each component is Gaussian:

$$P(r_k|s_{mk}) = \frac{1}{\sqrt{\pi N_0}} \exp \left[-\frac{(r_k - s_{mk})^2}{N_0} \right]$$

Therefore,

$$P(\mathbf{r}|s_m) = \left(\frac{1}{\sqrt{\pi N_0}} \right)^N \exp \left[-\frac{1}{N_0} \sum_{k=1}^N (r_k - s_{mk})^2 \right]$$

The receiver computes $P(\mathbf{r}|s_m)$ for all $m = 1, 2, \dots, M$ and chooses the largest value. Taking logarithm,

$$\ln P(\mathbf{r}|s_m) = -\frac{1}{N_0} \sum_{k=1}^N (r_k - s_{mk})^2 + \text{constant}$$

Hence, maximizing likelihood is equivalent to minimizing

$$D(\mathbf{r}, \mathbf{s}_m) = \sum_{k=1}^N (r_k - s_{mk})^2$$

which is the Euclidean distance.

Correlation Receiver Form

Expand the distance metric:

$$\begin{aligned} D(\mathbf{r}, \mathbf{s}_m) &= \sum_{k=1}^N (r_k^2 - 2r_k s_{mk} + s_{mk}^2) \\ &= \sum_{k=1}^N r_k^2 - 2 \sum_{k=1}^N r_k s_{mk} + \sum_{k=1}^N s_{mk}^2 \end{aligned}$$

The first term is independent of m , therefore it can be ignored. Hence,

$$\min_m D(\mathbf{r}, \mathbf{s}_m) \equiv \min_m \left[-2 \sum_{k=1}^N r_k s_{mk} + E_m \right]$$

where

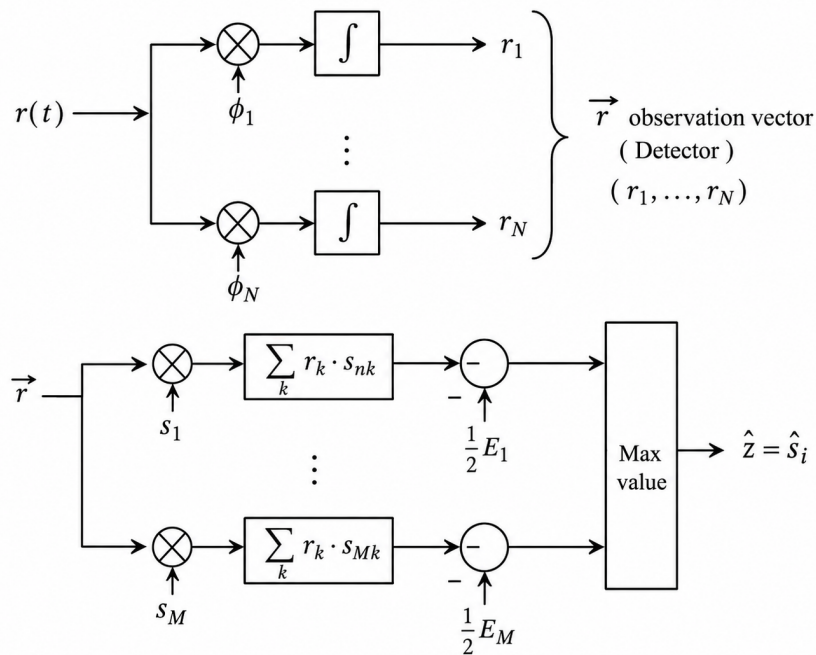
$$E_m = \sum_{k=1}^N s_{mk}^2$$

Multiplying by $-\frac{1}{2}$,

$$\max_m \left[\sum_{k=1}^N r_k s_{mk} - \frac{E_m}{2} \right]$$

Therefore, the correlation receiver chooses

$$\boxed{\max_m C(\mathbf{r}, \mathbf{s}_m) = \sum_{k=1}^N r_k s_{mk} - \frac{1}{2} E_m}$$



P_e for Bandpass Signals

1. BPSK ($k = 1, M = 2$)

$$P_e = \frac{1}{2} \operatorname{erfc} \left(\frac{d_{\min}}{2\sqrt{N_0}} \right) \quad (110)$$

$$d_{\min} = 2\sqrt{E_b} = 2\sqrt{E_s} \quad (111)$$

$$P_e = \frac{1}{2} \operatorname{erfc} \left(\frac{2\sqrt{E_b}}{2\sqrt{N_0}} \right) = \frac{1}{2} \operatorname{erfc} \left(\sqrt{\frac{E_b}{N_0}} \right) = \frac{1}{2} \operatorname{erfc} \left(\sqrt{\frac{E_s}{N_0}} \right) \quad (112)$$

2. QPSK ($k = 2, M = 4$)

$$d_{\min} = \sqrt{2E_s} \quad E_s = 2E_b \quad (113)$$

$$P_e = \frac{1}{2} \operatorname{erfc} \left(\sqrt{\frac{E_s}{2N_0}} \right) = \frac{1}{2} \operatorname{erfc} \left(\sqrt{\frac{E_b}{N_0}} \right) \quad (114)$$

In terms of E_b , the probability of error for QPSK and BPSK are the same.

3. M-PSK

$$d_{\min} = 2\sqrt{E_s} \sin \left(\frac{\pi}{M} \right) \quad (115)$$

Optimistic P_e : In M-PSK, each symbol has two immediate neighbors. Consider only the nearest neighbors for error calculations.

$$P_e = \frac{1}{M} 2(M-1) \left[\frac{1}{2} \operatorname{erfc} \left(\frac{d_{\min}}{2\sqrt{N_0}} \right) \right] \quad (116)$$

$$P_e \approx \operatorname{erfc} \left(\frac{2\sqrt{E_s} \sin(\pi/M)}{2\sqrt{N_0}} \right) = \operatorname{erfc} \left(\sqrt{\frac{E_s}{N_0}} \sin \left(\frac{\pi}{M} \right) \right) \quad (117)$$

$$E_s = kE_b = E_b \log_2 M \quad (118)$$

$$P_e \approx \operatorname{erfc} \left[\sqrt{\sin^2 \left(\frac{\pi}{M} \right) \frac{E_b \log_2 M}{N_0}} \right] \quad (119)$$

Pessimistic P_e : Consider all $(M - 1)$ symbols as adjacent symbols for error calculation.

$$P_e = M \cdot \frac{1}{M} (M - 1) \cdot \frac{1}{2} \operatorname{erfc} \left(\frac{d_{\min}}{2\sqrt{N_0}} \right) \quad (120)$$

$$P_e = \frac{M - 1}{2} \operatorname{erfc} \left[\sqrt{\sin^2 \left(\frac{\pi}{M} \right) \frac{E_b \log_2 M}{N_0}} \right] \quad (121)$$

4. M-PAM

Inner symbols: $(M - 2)$

Outer symbols: 2

Optimistic P_e for M-PAM

$$P_e = \frac{1}{M} \left[(M - 2) \cdot 2 \cdot \frac{1}{2} \operatorname{erfc} \left(\frac{d_{\min}}{2\sqrt{N_0}} \right) + 2 \cdot 1 \cdot \frac{1}{2} \operatorname{erfc} \left(\frac{d_{\min}}{2\sqrt{N_0}} \right) \right] \quad (122)$$

$$P_e = \frac{M - 1}{M} \operatorname{erfc} \left(\frac{d_{\min}}{2\sqrt{N_0}} \right) \quad (123)$$

Recall for M-PAM:

$$d_{\min} = \sqrt{\frac{12 \log_2 M}{M^2 - 1} E_{b,\text{avg}}} \quad (124)$$

$$P_e = \frac{M - 1}{M} \operatorname{erfc} \left(\sqrt{\frac{3 \log_2 M}{M^2 - 1} \frac{E_{b,\text{avg}}}{N_0}} \right) \quad (125)$$

Pessimistic P_e for M-PAM

$$P_e \approx \frac{M}{M} (M - 1) \frac{1}{2} \operatorname{erfc} \left(\sqrt{\frac{3 \log_2 M}{M^2 - 1} \frac{E_{b,\text{avg}}}{N_0}} \right) \quad (126)$$

5. M-QAM

$$\text{M-QAM} = \sqrt{M}\text{-PAM} \times \sqrt{M}\text{-PAM} \quad (127)$$

Optimistic P_e :

$$P_{c,\text{QAM}} = P_{c,\sqrt{M}\text{-PAM}} \cdot P_{c,\sqrt{M}\text{-PAM}} \quad (128)$$

$$P_{e,\text{QAM}} = 1 - P_{c,\text{QAM}} = 1 - [(1 - P_{e,\sqrt{M}\text{-PAM}})(1 - P_{e,\sqrt{M}\text{-PAM}})] \quad (129)$$

Let

$$x = P_{e,\sqrt{M}\text{-PAM}} \quad (130)$$

$$P_{e,\text{QAM}} = 1 - (1 - x)^2 = 1 - (1 + x^2 - 2x) \approx 2x \quad (131)$$

$$P_{e,\text{QAM}} \approx 2P_{e,\sqrt{M}\text{-PAM}} \quad (132)$$

$$P_{e,\text{QAM}} = 2 \left(1 - \frac{1}{\sqrt{M}} \right) \text{erfc} \left(\sqrt{\frac{3 \log_2 \sqrt{M} E_{b\text{avg}}}{M-1} \frac{1}{N_0}} \right) \quad (133)$$

Pessimistic P_e for QAM:

$$P_e \approx (\sqrt{M} - 1) \text{erfc} \left(\sqrt{\frac{3 \log_2 \sqrt{M} E_{b\text{avg}}}{M-1} \frac{1}{N_0}} \right) \quad (134)$$

6. BFSK

Coherent detection:

$$P_e = \frac{1}{2} \text{erfc} \left(\sqrt{\frac{E_b}{2N_0}} \right) \quad (135)$$

Noncoherent detection:

$$P_e \approx \frac{1}{2} \exp \left(-\frac{E_b}{2N_0} \right) \quad (136)$$

7. DPSK

$$P_e = \frac{1}{2} \exp \left(-\frac{E_b}{N_0} \right) \quad (137)$$

BER Formula Sheet (AWGN)

Useful relations

$$Q(x) = \frac{1}{2} \text{erfc} \left(\frac{x}{\sqrt{2}} \right), \quad \text{erfc}(x) = 2Q(\sqrt{2}x) \quad (138)$$

$$\text{erfc}(x) \approx \frac{e^{-x^2}}{\sqrt{\pi}}, \quad x > 0 \quad (139)$$

Baseband Binary

$$\text{Unipolar: } P_e = \frac{1}{2} \operatorname{erfc} \left(\sqrt{\frac{E_b}{4N_0}} \right) \quad (140)$$

$$\text{Polar/Manchester: } P_e = \frac{1}{2} \operatorname{erfc} \left(\sqrt{\frac{E_b}{N_0}} \right) \quad (141)$$

Binary Passband

$$\text{BPSK: } P_e = Q \left(\sqrt{\frac{2E_b}{N_0}} \right) = \frac{1}{2} \operatorname{erfc} \left(\sqrt{\frac{E_b}{N_0}} \right) \quad (142)$$

$$\text{Coherent BFSK: } P_e = Q \left(\sqrt{\frac{E_b}{N_0}} \right) \quad (143)$$

$$\text{Noncoherent BFSK: } P_e = \frac{1}{2} e^{-E_b/(2N_0)} \quad (144)$$

$$\text{DPSK: } P_e = \frac{1}{2} e^{-E_b/N_0} \quad (145)$$

QPSK

$$P_e = Q \left(\sqrt{\frac{2E_b}{N_0}} \right) \quad (146)$$

M-ary Schemes

$$\text{M-PSK: } P_e \approx \operatorname{erfc} \left(\sqrt{\frac{E_b \log_2 M}{N_0}} \sin \frac{\pi}{M} \right) \quad (147)$$

$$\text{M-PAM: } P_e = \frac{M-1}{M} \operatorname{erfc} \left(\sqrt{\frac{3 \log_2 M}{M^2 - 1} \frac{E_b}{N_0}} \right) \quad (148)$$

$$\text{Square M-QAM: } P_e \approx 2 \left(1 - \frac{1}{\sqrt{M}} \right) \operatorname{erfc} \left(\sqrt{\frac{3 \log_2 M}{M - 1} \frac{E_b}{N_0}} \right) \quad (149)$$

Numerical Examples**Example 1:**

Consider the octal signal point constellations in Figure 6.

1. The nearest-neighbor signal points in the 8-QAM signal constellation are separated in distance by A units. Determine the radii a and b of the inner and outer circles, respectively.
2. The adjacent signal points in the 8-PSK are separated by a distance of A units. Determine the radius r of the circle.
3. Determine the average transmitter powers for the two signal constellations, and compare the two powers. What is the relative power advantage of one constellation over the other? (Assume that all signal points are equally probable.)

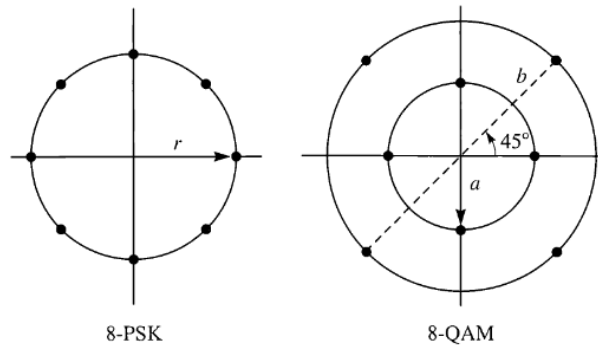


Figure 6: 8-PSK and 8-QAM Constellation Diagrams

Solution

1. Radius of 8-PSK constellation

In 8-PSK, adjacent symbols are separated by angle

$$\theta = \frac{2\pi}{8} = \frac{\pi}{4}.$$

Distance between adjacent points is given as A .

Using the cosine rule for the triangle formed by two adjacent points:

$$A^2 = r^2 + r^2 - 2r^2 \cos \theta$$

$$A^2 = 2r^2(1 - \cos \theta)$$

$$A^2 = 2r^2 \left(1 - \frac{1}{\sqrt{2}}\right)$$

$$r = \frac{A}{\sqrt{2 - \sqrt{2}}}$$

$$r \approx 1.306A$$

Average energy for 8-PSK:

$$E_{\text{avg,PSK}} = r^2 = (1.306A)^2 = 1.707A^2$$

—

2. Radii of 8-QAM constellation

Nearest neighbor distance is A .

Inner radius a

From geometry:

$$A = \sqrt{2}a$$

$$a = \frac{A}{\sqrt{2}} = 0.707A$$

—

Outer radius b

Using cosine rule between adjacent inner and outer points (separated by 45°):

$$A^2 = a^2 + b^2 - 2ab \cos 45^\circ$$

$$A^2 = a^2 + b^2 - 2ab \left(\frac{1}{\sqrt{2}} \right)$$

Substitute $a = \frac{A}{\sqrt{2}}$:

$$A^2 = \frac{A^2}{2} + b^2 \frac{2A}{\sqrt{2}} b \cdot \frac{1}{\sqrt{2}}$$

$$A^2 = \frac{A^2}{2} + b^2 - Ab$$

$$b^2 - Ab - \frac{A^2}{2} = 0$$

Solving:

$$b = \frac{A}{2}(1 + \sqrt{3})$$

$$b \approx 1.366A$$

—

Average energy of 8-QAM

There are four inner points and four outer points:

$$E_{\text{avg,QAM}} = \frac{1}{8}(4a^2 + 4b^2)$$

$$E_{\text{avg,QAM}} = \frac{1}{2}(a^2 + b^2)$$

$$E_{\text{avg,QAM}} = \frac{1}{2} \left[\left(\frac{A}{\sqrt{2}} \right)^2 + (1.366A)^2 \right]$$

$$E_{\text{avg,QAM}} \approx 1.183A^2$$

—

3. Power advantage

$$\begin{aligned}
 \text{Power advantage} &= 10 \log_{10} \left(\frac{E_{\text{avg,PSK}}}{E_{\text{avg,QAM}}} \right) \\
 &= 10 \log_{10} \left(\frac{1.707A^2}{1.183A^2} \right) \\
 &\approx 1.59 \text{ dB}
 \end{aligned}$$

8-QAM is more power efficient than 8-PSK by 1.59 dB

8-PSK

Optimistic probability of error:

$$P_e = p(s_i) \times (\text{number of symbols}) \times (\text{immediate neighbours}) \times \frac{1}{2} \text{erfc} \left(\frac{d_{\min}}{2\sqrt{N_0}} \right)$$

$$P_e = \frac{1}{8} \times 8 \times 2 \times \frac{1}{2} \text{erfc} \left(\frac{A}{2\sqrt{N_0}} \right)$$

$$P_e = \text{erfc} \left(\frac{A}{2\sqrt{N_0}} \right)$$

Pessimistic probability of error:

$$P_e = p(s_i) \times (\text{number of symbols}) \times (M - 1) \times \frac{1}{2} \text{erfc} \left(\frac{d_{\min}}{2\sqrt{N_0}} \right)$$

$$P_e = \frac{8}{8} \times 7 \times \frac{1}{2} \text{erfc} \left(\frac{A}{2\sqrt{N_0}} \right)$$

$$P_e = 3.5 \text{erfc} \left(\frac{A}{2\sqrt{N_0}} \right)$$

8-QAM*Optimistic probability of error:*

$$P_e = \frac{1}{8} [(4 \times 2) + (4 \times 4)] \frac{1}{2} \operatorname{erfc} \left(\frac{A}{2\sqrt{N_0}} \right)$$

$$P_e = 1.5 \operatorname{erfc} \left(\frac{A}{2\sqrt{N_0}} \right)$$

Pessimistic probability of error:

$$P_e = 3.5 \operatorname{erfc} \left(\frac{A}{2\sqrt{N_0}} \right)$$

Example 2:

Consider the two 8-point QAM signal constellations shown in Figure 7. The minimum distance between adjacent points is $2A$. Determine the average transmitted power for each constellation, assuming that the signal points are equally probable. Which constellation is more power-efficient?

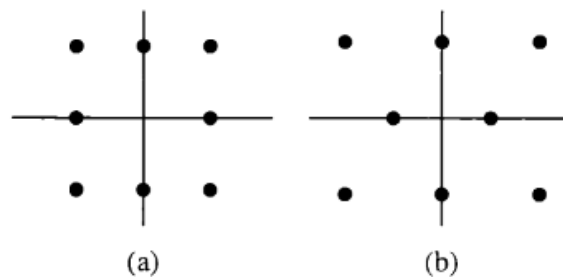


Figure 7: 8-QAM constellations

Figure 7a:

$$\begin{aligned} E_{\text{avg}} &= \frac{1}{8} [4(2A)^2 + 4(2\sqrt{2}A)^2] \\ &= \frac{1}{8} [4(4A^2) + 4(8A^2)] = \frac{1}{8} (16A^2 + 32A^2) = 6A^2 \end{aligned}$$

Figure 7b:

$$\begin{aligned} E_{\text{avg}} &= \frac{1}{8} [2A^2 + 2(A\sqrt{3})^2 + 4(A\sqrt{7})^2] \\ &= \frac{1}{8} (2A^2 + 6A^2 + 28A^2) = 4.5A^2 \end{aligned}$$

Hence, Figure 7b(8-QAM) has lower average energy than Figure 5a.

Figure 7a: Probability of error

Optimistic:

$$P_e = \frac{1}{8}[4 \times 2 + 2 \times 2 + 2 \times 2] \cdot \frac{1}{2} \operatorname{erfc}\left(\frac{2A}{2\sqrt{N_0}}\right)$$

$$P_e = \operatorname{erfc}\left(\frac{A}{\sqrt{N_0}}\right)$$

Pessimistic:

$$P_e = 3.5 \operatorname{erfc}\left(\frac{A}{\sqrt{N_0}}\right)$$

Figure 7b: Probability of error

Optimistic:

$$P_e = \frac{1}{8}[4 \times 2 + 2 \times 4 + 2 \times 5] \cdot \frac{1}{2} \operatorname{erfc}\left(\frac{2A}{2\sqrt{N_0}}\right)$$

$$P_e = 1.625 \operatorname{erfc}\left(\frac{A}{\sqrt{N_0}}\right)$$

Pessimistic:

$$P_e = 3.5 \operatorname{erfc}\left(\frac{A}{\sqrt{N_0}}\right)$$

Example 3:

The signal constellation for a communication system with 16 equiprobable symbols is shown in Figure 8. The channel is AWGN with noise power spectral density of $N_0/2$.

1. Using the union bound, find a bound in terms of A and N_0 on the error probability for this channel.
2. Determine the average SNR per bit for this channel.
3. Express the bound found in part 1 in terms of the average SNR per bit.
4. Compare the power efficiency of this system with a 16-level PAM system.

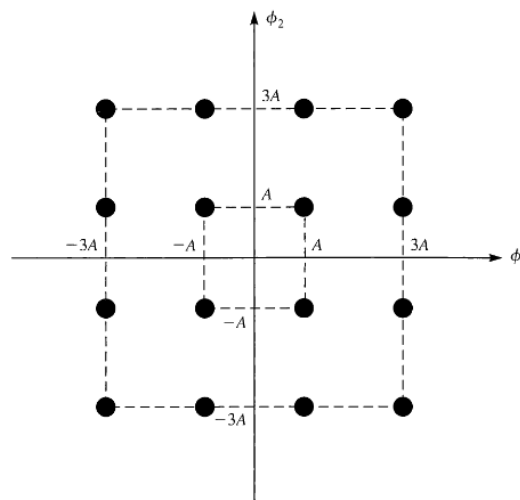


Figure 8: 16-QAM Constellations

Solution:

$$E_{\text{avg}} = E_{\text{savg}} = \frac{1}{16} \left[4(\sqrt{2}A)^2 + 8(\sqrt{10}A)^2 + 4(\sqrt{18}A)^2 \right] \quad (150)$$

$$= 10A^2 \quad (151)$$

$$E_{\text{bavg}} = \frac{E_{\text{savg}}}{\log_2 M} = \frac{10A^2}{4} = 2.5A^2 \quad (152)$$

$$A = \sqrt{\frac{E_{\text{bavg}}}{2.5}} \quad (153)$$

16-QAM (Optimistic bound):

$$P_e = \frac{1}{16} [4 \times 4 + 4 \times 2 + 8 \times 3] \frac{1}{2} \text{erfc} \left(\frac{2A}{2\sqrt{N_0}} \right) \quad (154)$$

$$= 3 \text{erfc} \left(\frac{A}{\sqrt{N_0}} \right) \quad (155)$$

$$= 3 \text{erfc} \left(\sqrt{\frac{E_{\text{bavg}}}{2.5N_0}} \right) \quad (156)$$

$$= 3 \text{erfc} \left(\sqrt{\frac{0.4E_{\text{bavg}}}{N_0}} \right) \quad (157)$$

16-PAM (Optimistic bound):

$$P_e = \text{erfc} \left[\sqrt{\frac{3 \log_2 M}{(M^2 - 1)} \frac{E_{\text{bavg}}}{N_0}} \right] \quad (158)$$

$$= \text{erfc} \left[\sqrt{0.047 \frac{E_{\text{bavg}}}{N_0}} \right] \quad (159)$$

Power efficiency comparison = η

$$\frac{\eta_{\text{QAM}}}{\eta_{\text{PAM}}} = \frac{0.4}{0.047} = 8.5 \approx 9.3 \text{ dB} \quad (160)$$

Conclusion: QAM is more power efficient.

16-QAM (Pessimistic bound):

$$P_e = 7.5 \text{erfc} \left(\frac{A}{\sqrt{N_0}} \right) \quad (161)$$

Example 4:

Consider two 8-QAM constellations shown in Figure 9. The Euclidean distance between any two adjacent points is A . Find the average symbol energy of the constellations in terms of A . Which is a more power efficient constellation and why? Obtain the optimistic union bounded P_e for the constellations.

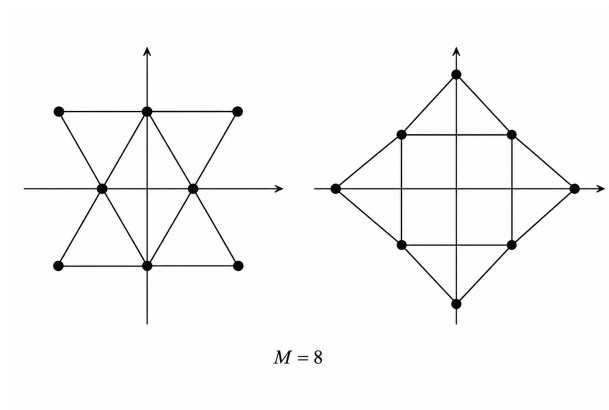


Figure 9: 8-QAM Constellations

Fig 9a:8-QAM

$$E_{avg} = \frac{1}{8} \left[2 \left(\frac{A}{2} \right)^2 + 2 \left(\frac{\sqrt{3}}{2} A \right)^2 + 4 \left(\frac{\sqrt{7}}{2} A \right)^2 \right] = 1.125 A^2$$

Optimistic

$$P_e = \frac{1}{8} [4 \times 2 + 2 \times 5 + 2 \times 4] \frac{1}{2} \operatorname{erfc} \left(\frac{A}{2\sqrt{N_0}} \right)$$

$$= 1.625 \operatorname{erfc} \left(\frac{A}{2\sqrt{N_0}} \right)$$

Pessimistic

$$P_e = \frac{1}{8} \times 8 \times 7 \times \frac{1}{2} \operatorname{erfc} \left(\frac{A}{2\sqrt{N_0}} \right) = 3.5 \operatorname{erfc} \left(\frac{A}{2\sqrt{N_0}} \right)$$

Fig 9b: 8-QAM

$$E_{s,avg} = \frac{1}{8} \left[4 \left(\frac{A}{\sqrt{2}} \right)^2 + 4 \left(\frac{1+\sqrt{3}}{2} A \right)^2 \right] = 1.183 A^2$$

optimistic

$$P_e = \frac{1}{8} [4 \times 2 + 4 \times 4] \frac{1}{2} \operatorname{erfc} \left(\frac{A}{2\sqrt{N_0}} \right)$$

$$= 1.5 \operatorname{erfc} \left(\frac{A}{2\sqrt{N_0}} \right)$$

Pessimistic

$$P_e = 3.5 \operatorname{erfc} \left(\frac{A}{2\sqrt{N_0}} \right)$$

Example 5:

Consider the 16-QAM signal constellation shown in the figure 10. The distance between any two adjacent points is A . Express the signal in terms of its basis functions. Find the average energy of the constellation and obtain expressions for a union bounded probability of error.

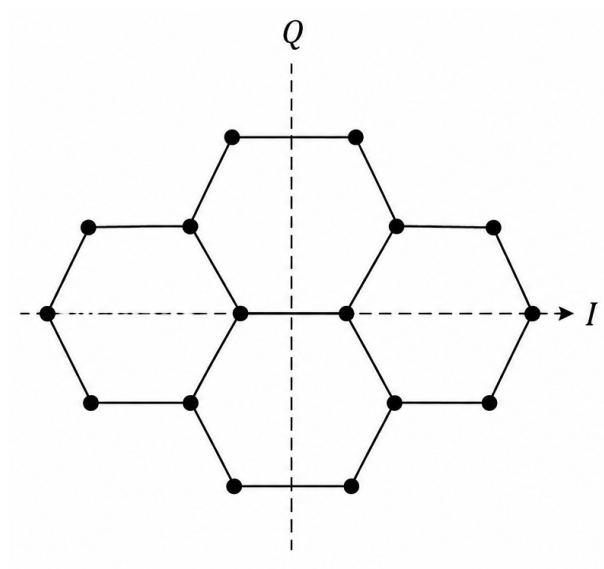


Figure 10: 16-QAM Constellation

Solution:

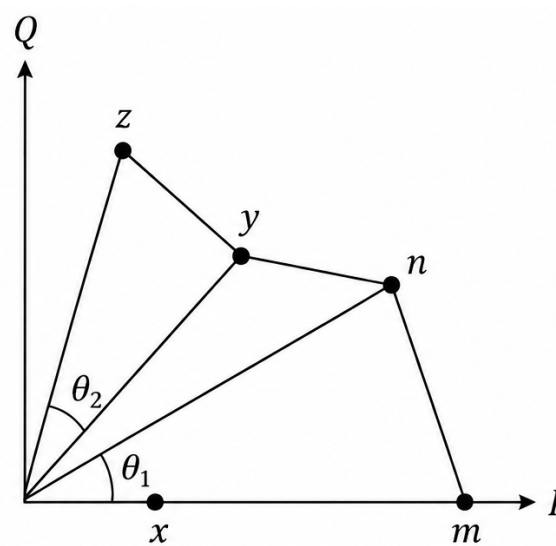


Figure 11: First Quadrant

$$x = \frac{A}{2}$$

$$y^2 = x^2 + A^2 - 2Ax \cos(120^\circ)$$

$$y = \frac{\sqrt{7}}{2}A$$

$$z^2 = (\sqrt{3}A) + \left(\frac{A}{2}\right)^2 = \frac{\sqrt{13}}{2}A$$

$$m = 2A + \frac{A}{2} = \frac{5}{2}A$$

$$n = \left(\frac{\sqrt{3}}{2} A \right)^2 + (2A)^2 = \frac{\sqrt{19}}{2} A$$

$$E_{\text{avg}} = \frac{1}{16} [2x^2 + 4y^2 + 4z^2 + 2m^2 + 4n^2]$$

$$E_{\text{avg}} = 3.25A^2$$

Optimistic

$$P_e = \frac{1}{16} [2 \times 2 + 8 \times 2 + 6 \times 3] \frac{1}{2} \text{erfc} \left(\frac{A}{2\sqrt{N_0}} \right)$$

$$P_e = 1.1875 \text{erfc} \left(\frac{A}{2\sqrt{N_0}} \right)$$

Pessimistic

$$P_e = 7.5 \text{erfc} \left(\frac{A}{2\sqrt{N_0}} \right)$$

Example 6:

A binary PSK is used for a transmission over AWGN with a PSD of $\frac{1}{2}N_0 = 10^{-10}$ W/Hz.

The transmitted signal energy $E_b = \frac{1}{2}A^2T$, where T is the bit interval and A is the signal amplitude. Determine the signal amplitude required to achieve $P_e = 10^{-6}$ when the data rate is 10 kbps, 100 kbps and 1 Mbps.

Solution:

$$P_e = \frac{1}{2} \text{erfc} \left(\sqrt{\frac{E_b}{N_0}} \right)$$

$$10^{-6} = \frac{1}{2} \text{erfc} \left(\sqrt{\frac{A^2T}{2N_0}} \right)$$

Using the approximation

$$\text{erfc}(x) \approx \frac{e^{-x^2}}{\sqrt{\pi}}$$

$$\ln(2x\sqrt{\pi} 10^{-6}) = \frac{A^2T}{2N_0}$$

$$12.54 = \frac{A^2T}{2N_0}$$

$$A = \sqrt{12.54 \cdot 2N_0R}$$

For different data rates:

$$R = 10 \text{ kbps}, \quad A = 7.08 \text{ mV}$$

$$R = 100 \text{ kbps}, \quad A = 22.39 \text{ mV}$$

$$R = 1 \text{ Mbps}, \quad A = 70.82 \text{ mV}$$

Example 7:

Consider a 4-PSK and 8-PSK shown in Figure 12. Determine the radii r_1 and r_2 of the circles such that the distance between two points is d . Determine the additional transmitted energy required in the 8-PSK signal to achieve the same P_e as the 4-PSK signal at high SNR, where the P_e is determined by optimistic method.

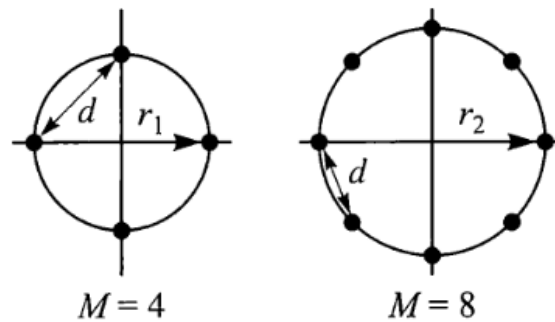


Figure 12: QPSK and 8PSK Constellations

Solution

For 4-PSK:

$$r_1^2 + r_1^2 = d^2$$

$$2r_1^2 = d^2$$

$$r_1 = \frac{d}{\sqrt{2}}$$

Average energy for 4-PSK:

$$E_{\text{avg}} = 4 \cdot \frac{1}{4} r_1^2 = r_1^2 = \frac{d^2}{2}$$

For 8-PSK:

Using cosine rule for adjacent points separated by 45° :

$$d^2 = r_2^2 + r_2^2 - 2r_2^2 \cos 45^\circ$$

$$d^2 = 2r_2^2(1 - \cos 45^\circ)$$

$$d^2 = 2r_2^2 \left(1 - \frac{1}{\sqrt{2}} \right)$$

$$r_2 = \frac{d^2}{\sqrt{2 - \sqrt{2}}}$$

Average energy for 8-PSK:

$$E_{\text{avg}} = 8 \cdot \frac{1}{8} r_2^2 = r_2^2 = \frac{d^2}{2 - \sqrt{2}}$$

Energy ratio:

$$\frac{E_{\text{avg},8\text{PSK}}}{E_{\text{avg},4\text{PSK}}} = \frac{\frac{d^2}{2 - \sqrt{2}}}{\frac{d^2}{2}} = \frac{2}{2 - \sqrt{2}} \approx 3.44$$

In decibels:

$$10 \log_{10}(3.44) \approx 5.33 \text{ dB}$$

$$E_{\text{avg},8\text{PSK}} \approx 3.44 E_{\text{avg},4\text{PSK}}$$

$$E_{\text{avg},8\text{PSK}} \approx E_{\text{avg},4\text{PSK}} + 5.33 \text{ dB}$$

Example 8:

Digital information is to be transmitted by carrier modulation through an AWGN channel with bandwidth $B = 100 \text{ kHz}$ and noise power spectral density $N_0 = 10^{-10} \text{ W/Hz}$. Determine the maximum data rate that can be transmitted through the channel for QPSK, BFSK, 4-FSK which is detected non-coherently

Solution:

1. QPSK

Bandwidth relation:

$$W = \frac{R_b}{\log_2 M}$$

For QPSK, $M = 4$, so:

$$\log_2 4 = 2$$

$$B = \frac{R_b}{2}$$

$$R_b = 2B$$

$$R_b = 2 \times 100 \text{ kHz}$$

$$R_b = 200 \text{ kbps}$$

2. BFSK

Bandwidth relation:

$$B = \frac{MR_b}{2 \log_2 M}$$

For BPSK, $M = 2$, so:

$$\log_2 2 = 1$$

$$B = \frac{2R_b}{2} = R_b$$

$$R_b = B = 100 \text{ kbps}$$

3. 4-FSK (Non-coherent)

Bandwidth relation:

$$B = \frac{MR_b}{2 \log_2 M}$$

For 4-FSK, $M = 4$, so:

$$\log_2 4 = 2$$

$$B = \frac{4R_b}{2 \times 2} = R_b$$

$$R_b = B = 100 \text{ kbps}$$

Example 9:

A binary valued signal $s(t)$ is to be transmitted as a one shot signal over a link with AWGN. The symbols are defined as

$$S_1(t) = \begin{cases} A, & 0 \leq t \leq T \\ 0, & \text{otherwise} \end{cases}$$

and

$$S_2(t) = \begin{cases} A, & 0 \leq t \leq \frac{T}{2} \\ -A, & \frac{T}{2} \leq t \leq T \\ 0, & \text{otherwise} \end{cases}$$

1. Sketch the impulse responses of the filters matched to $S_1(t)$ and $S_2(t)$.
2. Determine the outputs of the matched filters when $S_1(t)$ is transmitted.
3. Determine the outputs of the matched filters when $S_2(t)$ is transmitted.
4. Show that the matched filter outputs are equivalent to correlator outputs.

Solution:

Consider the two signals

$$s_1(t) = \begin{cases} A, & 0 \leq t \leq T \\ 0, & \text{otherwise} \end{cases}$$

and

$$s_2(t) = \begin{cases} A, & 0 \leq t \leq \frac{T}{2} \\ -A, & \frac{T}{2} \leq t \leq T \\ 0, & \text{otherwise} \end{cases}$$

(a) Matched Filter Impulse Responses

Matched filter response:

$$h_i(t) = s_i(T - t)$$

Hence,

$$h_1(t) = \begin{cases} A, & 0 \leq t \leq T \\ 0, & \text{otherwise} \end{cases}$$

and

$$h_2(t) = \begin{cases} -A, & 0 \leq t \leq \frac{T}{2} \\ A, & \frac{T}{2} \leq t \leq T \\ 0, & \text{otherwise} \end{cases}$$

(b) Matched Filter Outputs**Output of MF1**

$$y_1(t) = s_1(t) * h_1(t)$$

Since both are rectangular pulses,

$$y_1(t) = \begin{cases} A^2 t, & 0 \leq t \leq T \\ A^2 (2T - t), & T \leq t \leq 2T \\ 0, & \text{otherwise} \end{cases}$$

Peak output:

$$\boxed{y_1(T) = A^2 T}$$

Output of MF2

$$y_2(t) = s_2(t) * h_2(t)$$

The waveform contains positive and negative triangular sections.
Important sampled values:

$$y_2(0) = 0$$

$$y_2\left(\frac{T}{2}\right) = -\frac{A^2T}{2}$$

$$y_2(T) = A^2T$$

$$y_2\left(\frac{3T}{2}\right) = -\frac{A^2T}{2}$$

$$y_2(2T) = 0$$

Thus,

$$\boxed{y_2(T) = A^2T}$$

is the peak output.

(c) Correlator Outputs

For signal $s_1(t)$,

$$\int_0^T s_1^2(t) dt = \int_0^T A^2 dt = A^2T$$

For signal $s_2(t)$,

$$\begin{aligned} \int_0^T s_2^2(t) dt &= \int_0^{T/2} A^2 dt + \int_{T/2}^T (-A)^2 dt \\ &= A^2 \frac{T}{2} + A^2 \frac{T}{2} = A^2T \end{aligned}$$

Hence,

$$\boxed{\text{Matched filter peak output} = \text{Correlator output} = A^2T}$$

Example 10

Consider the two signals

$$s_1(t) = \begin{cases} \frac{A}{2}, & 0 \leq t \leq \frac{T}{2} \\ -\frac{A}{2}, & \frac{T}{2} \leq t \leq T \\ 0, & \text{otherwise} \end{cases}$$

and

$$s_2(t) = \begin{cases} -\frac{A}{2}, & 0 \leq t \leq \frac{T}{4} \\ \frac{A}{2}, & \frac{T}{4} \leq t \leq \frac{T}{2} \\ -\frac{A}{2}, & \frac{T}{2} \leq t \leq \frac{3T}{4} \\ \frac{A}{2}, & \frac{3T}{4} \leq t \leq T \\ 0, & \text{otherwise} \end{cases}$$

1. Obtain the matched filter impulse responses corresponding to $s_1(t)$ and $s_2(t)$.
2. Determine the outputs:

$$y_3(t) = s_2(t) * h_1(t)$$

and

$$y_4(t) = s_1(t) * h_2(t)$$

where $h_1(t)$ and $h_2(t)$ are the matched filters for $s_1(t)$ and $s_2(t)$, respectively.

3. Verify whether the signals $s_1(t)$ and $s_2(t)$ are orthogonal.
4. Show that the correlator outputs at $t = T$ are zero for orthogonal signals.

Solution:

Consider the signals

$$s_1(t) = \begin{cases} \frac{A}{2}, & 0 \leq t \leq \frac{T}{2} \\ -\frac{A}{2}, & \frac{T}{2} \leq t \leq T \\ 0, & \text{otherwise} \end{cases}$$

and

$$s_2(t) = \begin{cases} -\frac{A}{2}, & 0 \leq t \leq \frac{T}{4} \\ \frac{A}{2}, & \frac{T}{4} \leq t \leq \frac{2T}{4} \\ -\frac{A}{2}, & \frac{2T}{4} \leq t \leq \frac{3T}{4} \\ \frac{A}{2}, & \frac{3T}{4} \leq t \leq T \\ 0, & \text{otherwise} \end{cases}$$

Matched Filters

$$h_1(t) = s_1(T - t)$$

$$h_2(t) = s_2(T - t)$$

Cross Outputs

Output $y_3(t)$

$$y_3(t) = s_2(t) * h_1(t)$$

Since the signals are orthogonal,

$$\boxed{y_3(T) = 0}$$

The waveform consists of alternating positive and negative triangular sections.
Correct peak magnitudes are

$$\boxed{\pm \frac{A^2 T}{16}}$$

and NOT

$$\frac{3A^2 T}{16}$$

Output $y_4(t)$

$$y_4(t) = s_1(t) * h_2(t)$$

Similarly,

$$\boxed{y_4(T) = 0}$$

and the maximum magnitudes are

$$\boxed{\pm \frac{A^2 T}{16}}$$

Orthogonality Verification

$$\begin{aligned}
 & \int_0^T s_1(t) s_2(t) dt \\
 &= \int_0^{T/4} \left(\frac{A}{2}\right) \left(-\frac{A}{2}\right) dt + \int_{T/4}^{T/2} \left(\frac{A}{2}\right) \left(\frac{A}{2}\right) dt \\
 &+ \int_{T/2}^{3T/4} \left(-\frac{A}{2}\right) \left(-\frac{A}{2}\right) dt + \int_{3T/4}^T \left(-\frac{A}{2}\right) \left(\frac{A}{2}\right) dt \\
 &= -\frac{A^2 T}{16} + \frac{A^2 T}{16} + \frac{A^2 T}{16} - \frac{A^2 T}{16} \\
 &= 0
 \end{aligned}$$

Therefore,

$$s_1(t) \perp s_2(t)$$

Hence correlator outputs at $t = T$ are zero:

$$y_3(T) = y_4(T) = 0$$

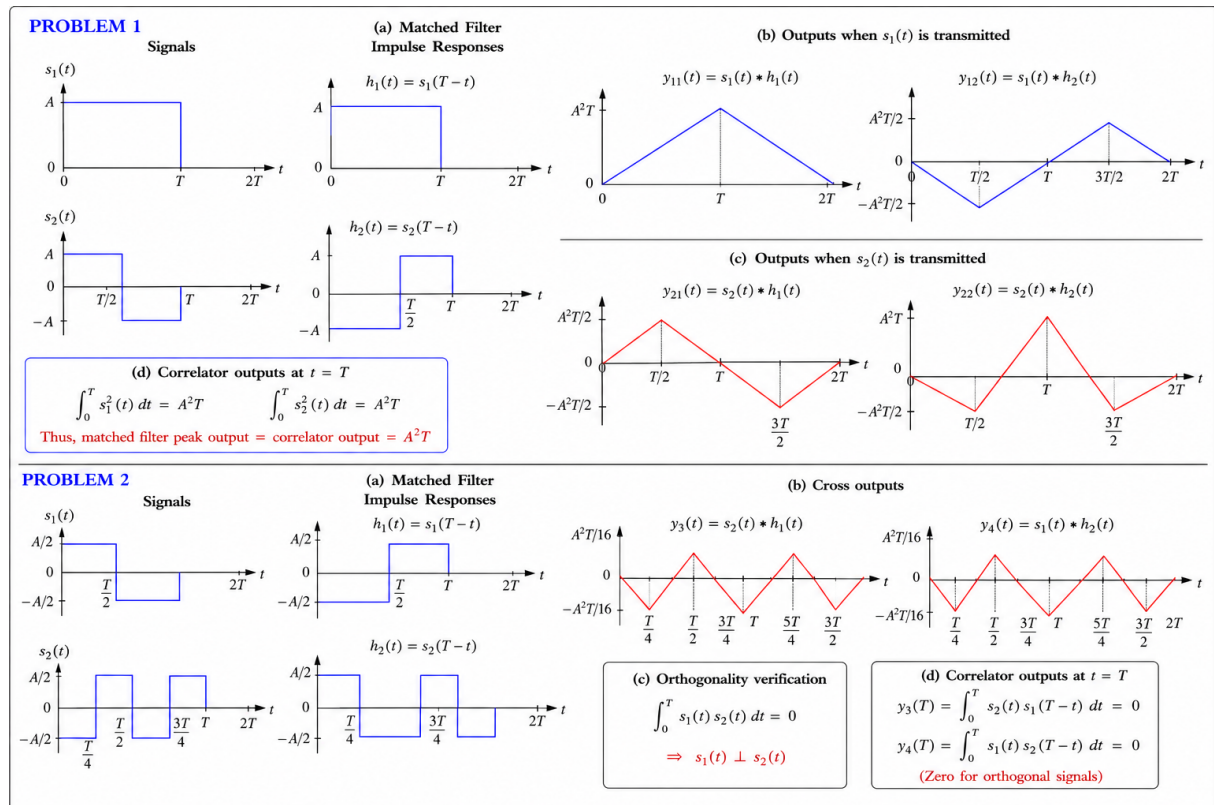


Figure 13: Waveforms for Example 9 and 10

Example 11:

A binary valued signal $s(t)$ is to be transmitted as a one shot signal over a link with AWGN. The symbols are defined as

$$S_1(t) = \begin{cases} A, & 0 \leq t \leq \frac{T}{2} \\ 0, & \text{otherwise} \end{cases}$$

and

$$S_2(t) = \begin{cases} A, & \frac{T}{2} \leq t \leq T \\ 0, & \text{otherwise} \end{cases}$$

Sketch the impulse response of the filters matched to $S_1(t)$ and $S_2(t)$. What is the output of the filters when $S_1(t)$ is sent. Also the outputs when $S_2(t)$ is sent. Demonstrate how one matched filter can be employed for detection of both signals.

Solution:

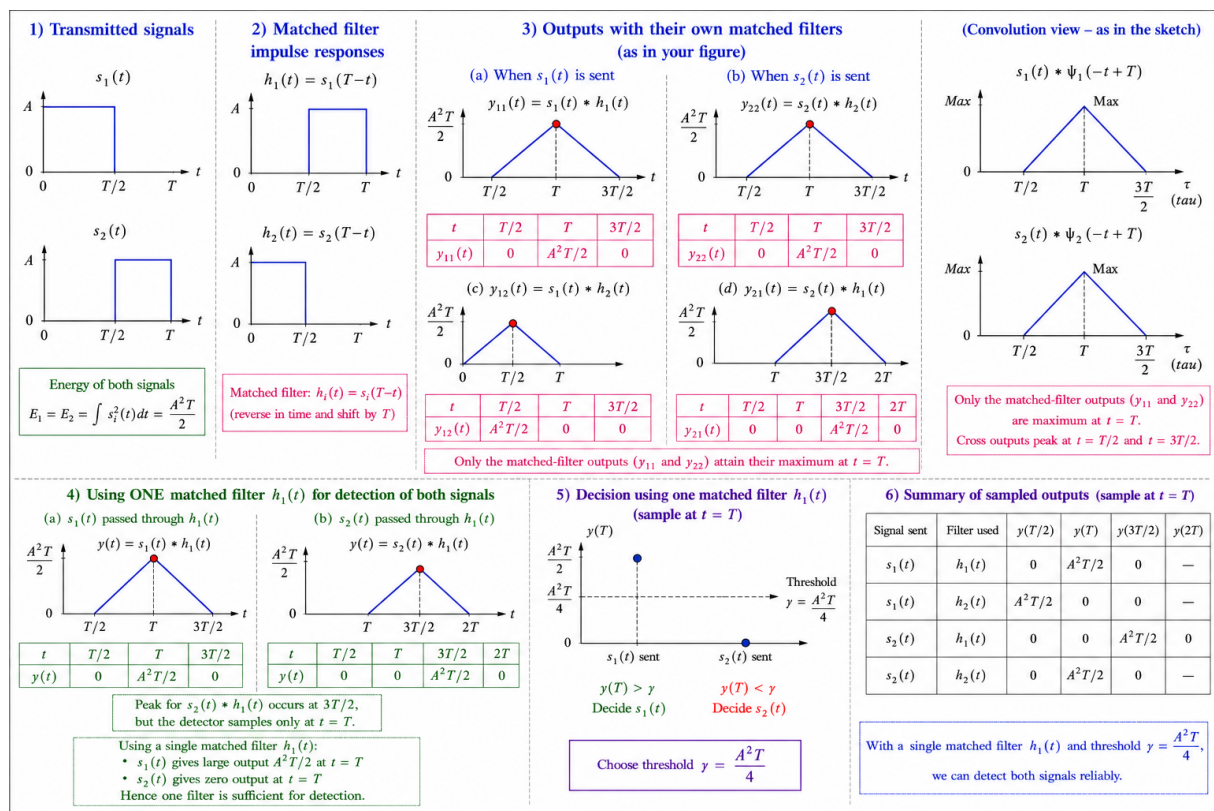


Figure 14: MF Analysis

Example 12:

A communication system transmits one of the three symbols $S_1(t)$, $S_2(t)$, and $S_3(t)$. The signal

$$S_1(t) = A \quad \text{over } [0, T]$$

$$S_2(t) = \begin{cases} A, & \text{over } [0, T/3] \\ -2A, & \text{over } [T/3, T] \end{cases}$$

and

$$S_3(t) = \begin{cases} -A, & \text{over } [0, T/3] \\ -2A, & \text{over } [T/3, T] \end{cases}$$

The channel is an AWGN channel with power spectral density equal to $N_0/2$ watts/Hz.

- Determine an orthonormal basis for this signal set, and depict the signal constellation.
- If the signals are equiprobable, what are the optimal decision rules for this system? Show the optimal decision regions on the signal constellation you plotted in part (a).
- If the signals are equiprobable, express the error probability of the optimal detector in terms of the average SNR per bit.

Solution:

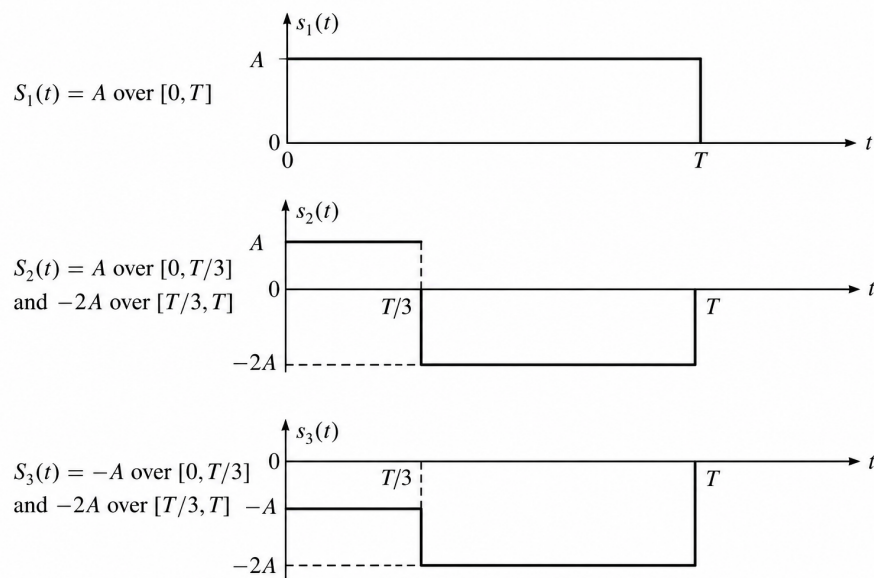


Figure 15: Signals

(a) Orthonormal Basis Functions

Since the signals change at

$$t = \frac{T}{3}$$

two orthonormal basis functions are sufficient.

Basis Function $\phi_1(t)$

$$\phi_1(t) = \begin{cases} \sqrt{\frac{3}{T}}, & 0 \leq t \leq \frac{T}{3} \\ 0, & \text{otherwise} \end{cases}$$

Verification:

$$\int_0^{T/3} \phi_1^2(t) dt = \frac{3}{T} \cdot \frac{T}{3} = 1$$

Basis Function $\phi_2(t)$

$$\phi_2(t) = \begin{cases} \sqrt{\frac{3}{2T}}, & \frac{T}{3} \leq t \leq T \\ 0, & \text{otherwise} \end{cases}$$

Verification:

$$\int_{T/3}^T \phi_2^2(t) dt = \frac{3}{2T} \cdot \frac{2T}{3} = 1$$

Also,

$$\int_0^T \phi_1(t)\phi_2(t) dt = 0$$

Hence,

$$\boxed{\{\phi_1(t), \phi_2(t)\}}$$

forms an orthonormal basis.

Signal Coordinates**Signal $s_1(t)$**

$$s_{11} = \int_0^{T/3} A\sqrt{\frac{3}{T}} dt = A\sqrt{\frac{T}{3}}$$

$$s_{12} = \int_{T/3}^T A\sqrt{\frac{3}{2T}} dt = A\sqrt{\frac{2T}{3}}$$

Therefore,

$$\boxed{s_1 = \left(A\sqrt{\frac{T}{3}}, A\sqrt{\frac{2T}{3}} \right)}$$

Signal $s_2(t)$

$$s_{21} = A\sqrt{\frac{T}{3}}$$

$$s_{22} = -2A\sqrt{\frac{2T}{3}}$$

Thus,

$$s_2 = \left(A\sqrt{\frac{T}{3}}, -2A\sqrt{\frac{2T}{3}} \right)$$

Signal $s_3(t)$

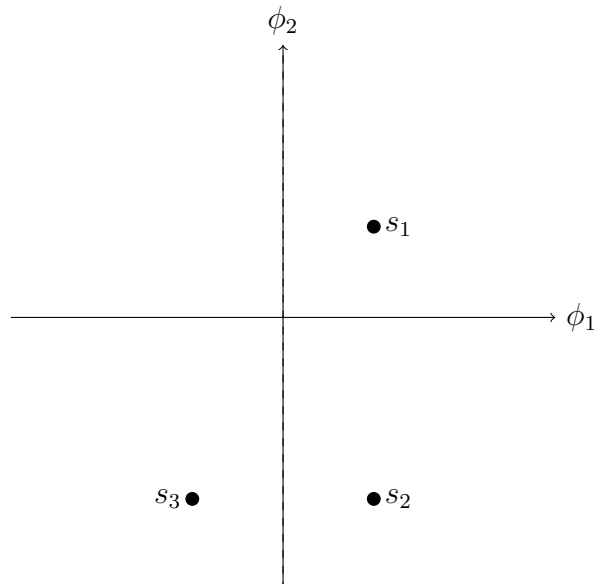
$$s_{31} = -A\sqrt{\frac{T}{3}}$$

$$s_{32} = -2A\sqrt{\frac{2T}{3}}$$

Hence,

$$s_3 = \left(-A\sqrt{\frac{T}{3}}, -2A\sqrt{\frac{2T}{3}} \right)$$

Signal Constellation



(b) Optimal Decision Rules

For equiprobable signals in AWGN, the optimum detector is the minimum-distance detector.

Choose the nearest constellation point

Decision Boundary Between s_2 and s_3

Since the two points differ only in the ϕ_1 coordinate,

$$\boxed{x = 0}$$

Decision Boundary Between s_1 and s_2

Midpoint:

$$\left(A\sqrt{\frac{T}{3}}, -\frac{1}{2}A\sqrt{\frac{2T}{3}} \right)$$

Boundary equation:

$$\boxed{y = -\frac{1}{2}A\sqrt{\frac{2T}{3}}}$$

Decision Boundary Between s_1 and s_3

Coordinates:

$$s_1 = \left(A\sqrt{\frac{T}{3}}, A\sqrt{\frac{2T}{3}} \right)$$

$$s_3 = \left(-A\sqrt{\frac{T}{3}}, -2A\sqrt{\frac{2T}{3}} \right)$$

Slope of line joining s_1 and s_3 :

$$m = \frac{A\sqrt{2T/3} + 2A\sqrt{2T/3}}{A\sqrt{T/3} + A\sqrt{T/3}} = \frac{3\sqrt{2}}{2}$$

Hence perpendicular slope:

$$m_{\perp} = -\frac{2}{3\sqrt{2}}$$

Midpoint:

$$\left(0, -\frac{1}{2}A\sqrt{\frac{2T}{3}} \right)$$

Therefore the perpendicular bisector is

$$\boxed{y = -\frac{2}{3\sqrt{2}}x - \frac{1}{2}A\sqrt{\frac{2T}{3}}}$$

Distance Between Signal Points

Distance d_{12}

$$\begin{aligned} d_{12}^2 &= (0)^2 + \left(3A\sqrt{\frac{2T}{3}}\right)^2 \\ &= 6A^2T \end{aligned}$$

Hence,

$$\boxed{d_{12} = \sqrt{6A^2T}}$$

Distance d_{23}

$$d_{23}^2 = \left(2A\sqrt{\frac{T}{3}}\right)^2 = \frac{4A^2T}{3}$$

Thus,

$$\boxed{d_{23} = 2A\sqrt{\frac{T}{3}}}$$

Distance d_{13}

$$\begin{aligned} d_{13}^2 &= \left(2A\sqrt{\frac{T}{3}}\right)^2 + \left(3A\sqrt{\frac{2T}{3}}\right)^2 \\ &= \frac{4A^2T}{3} + 6A^2T = \frac{22A^2T}{3} \end{aligned}$$

Therefore,

$$\boxed{d_{13} = \sqrt{\frac{22A^2T}{3}}}$$

Minimum Distance

$$\boxed{d_{\min} = d_{23} = 2A\sqrt{\frac{T}{3}}}$$

Hence dominant errors occur between

$$\boxed{s_2 \leftrightarrow s_3}$$

(c) Probability of Error

For equiprobable signals,

$$P_e = \frac{1}{3} \sum_{i=1}^3 P(e|s_i)$$

Using pairwise error approximation,

$$P_e \approx \frac{1}{3} \sum_{i=1}^3 \sum_{k \neq i} Q\left(\frac{d_{ik}}{\sqrt{2N_0}}\right)$$

where

$$Q(x) = \frac{1}{2} \operatorname{erfc}\left(\frac{x}{\sqrt{2}}\right)$$

Thus,

$$P_e \approx \frac{1}{3} \left[2Q\left(\frac{d_{12}}{\sqrt{2N_0}}\right) + 2Q\left(\frac{d_{13}}{\sqrt{2N_0}}\right) + 2Q\left(\frac{d_{23}}{\sqrt{2N_0}}\right) \right]$$

Since nearest-neighbor errors dominate,

$$P_e \approx \frac{2}{3} Q\left(\frac{2A\sqrt{T/3}}{\sqrt{2N_0}}\right)$$

Average Signal Energy

Energy of s_1

$$E_1 = A^2 T$$

Energy of s_2

$$E_2 = A^2 \frac{T}{3} + 4A^2 \frac{2T}{3} = 3A^2 T$$

Energy of s_3

$$E_3 = 3A^2 T$$

Hence average energy:

$$\begin{aligned} E_{\text{avg}} &= \frac{1}{3}(E_1 + E_2 + E_3) \\ &= \frac{1}{3}(A^2 T + 3A^2 T + 3A^2 T) \end{aligned}$$

$$= \frac{7A^2 T}{3}$$

Thus,

$$A^2 = \frac{3E_{\text{avg}}}{7T}$$

Probability of Error in Terms of Average Energy

Since

$$d_{\min} = 2A\sqrt{\frac{T}{3}}$$

substitute

$$A^2 = \frac{3E_{\text{avg}}}{7T}$$

Then

$$d_{\min} = 2\sqrt{\frac{E_{\text{avg}}}{7}}$$

Hence,

$$P_e \approx \frac{2}{3}Q\left(\sqrt{\frac{2E_{\text{avg}}}{7N_0}}\right)$$

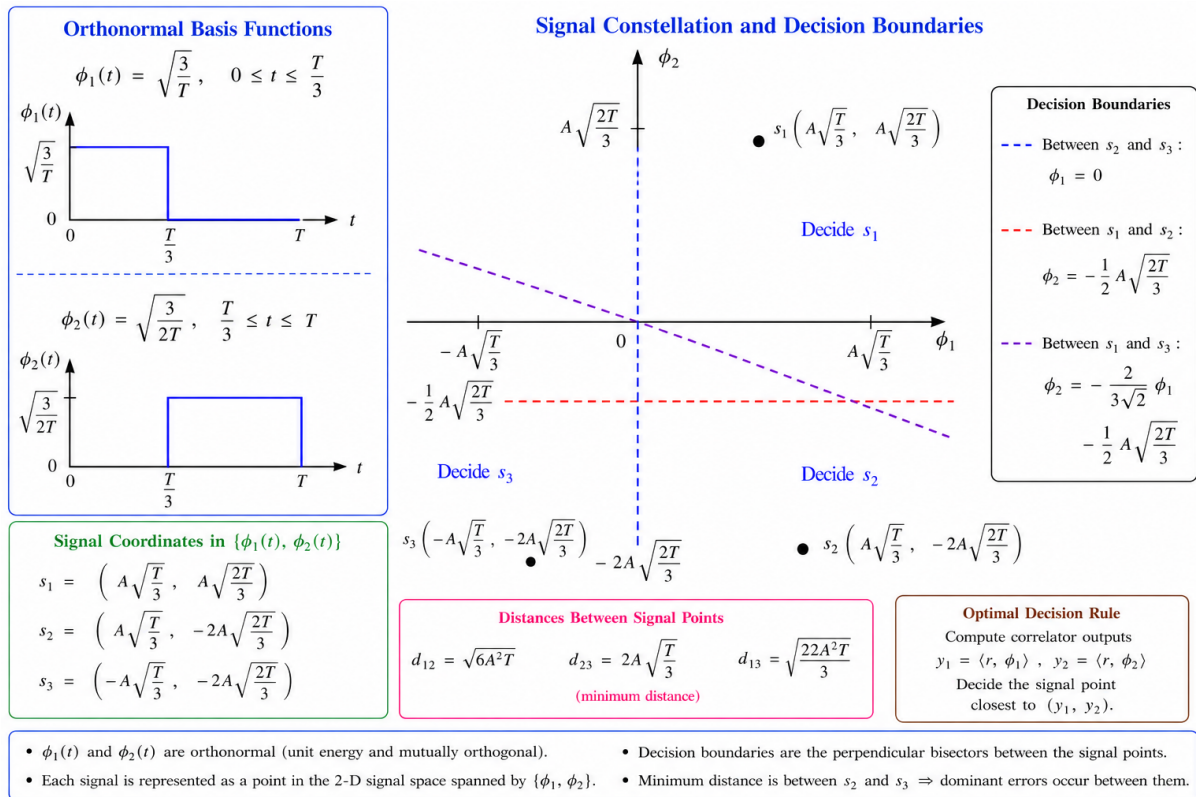


Figure 16: Constellation Diagram

BER Comparison of Signaling Schemes over AWGN Channels

Much of the material covered in this chapter has been devoted to digital modulation schemes operating over AWGN channels. In this section, we present a summary of the BERs of some popular digital modulation schemes, classified into two categories, depending on the method of detection used in the receiver:

Class I: Coherent detection

- Binary PSK: two symbols, single carrier
- Binary FSK: two symbols, two carriers, one for each symbol
- QPSK: four symbols, single carrier—the QPSK also includes the QAM, employing four symbols as a special case
- MSK: four symbols, two carriers

Class II: Noncoherent detection

- DPSK: two symbols, single carrier
- Binary FSK: two symbols, two carriers

Table 2 presents a summary of the formulas of the BERs of these schemes separated under Classes I and II. All the formulas are defined in terms of the ratio of energy per bit to the noise spectral density, E_b/N_0 , as summarized herein:

1. Under Class I, the formulas are expressed in terms of the Q -function. This function is defined as the area under the tail of the standard Gaussian distribution with zero mean and unit variance; the lower limit in the integral defining the Q -function is dependent solely on E_b/N_0 , scaled by the factor 2 for binary PSK, QPSK, and MSK. Naturally, as this SNR ratio is increased, the area under the Q -function is reduced and with it the BER is correspondingly reduced.
2. Under Class II, the formulas are expressed in terms of an exponential function, where the negative exponent depends on the E_b/N_0 ratio for DPSK and its scaled version by the factor 1/2 for binary FSK. Here again, as the E_b/N_0 is increased, the BER is correspondingly reduced.

Table 2: BER formulas for digital modulation schemes

Class	Signaling Scheme	BER
I. Coherent detection	Binary PSK, QPSK, MSK	$Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$
	Binary FSK	$Q\left(\sqrt{\frac{E_b}{N_0}}\right)$
II. Noncoherent detection	DPSK	$\frac{1}{2} \exp\left(-\frac{E_b}{N_0}\right)$
	Binary FSK	$\frac{1}{2} \exp\left(-\frac{E_b}{2N_0}\right)$

The performance curves of the digital modulation schemes listed in Table 2 are shown in Figure 36 where the BER is plotted versus E_b/N_0 . As expected, the BERs for all the schemes decrease monotonically with increasing E_b/N_0 , with all the graphs having a similar shape in the form of a waterfall.

Moreover, we can make the following observations:

1. For any value of E_b/N_0 , the schemes using coherent detection produce a smaller BER than those using noncoherent detection.
2. PSK schemes employing two symbols, namely binary PSK with coherent detection and DPSK with noncoherent detection, require an E_b/N_0 that is 3 dB less than their FSK counterparts to realize the same BER.
3. At high values of E_b/N_0 , DPSK and binary FSK using noncoherent detection perform almost as well, to within about 1 dB of their respective counterparts using coherent detection for the same BER.
4. Although under Class I the BER for binary PSK, QPSK, and MSK is governed by the same formula, there are important differences between them:
 - For the same channel bandwidth and BER, QPSK accommodates the transmission of binary data at twice the rate attainable with binary PSK; in other words, QPSK is bandwidth conserving.
 - When sensitivity to interfering signals is an issue of practical concern, as in wireless communications, MSK is preferred over QPSK.

Note: Conversion formulas and approximate equations for P_e

Conversion Between Q -Function and erfcFunction

Definitions

Q -function:

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-t^2/2} dt \quad (162)$$

Complementary error function:

$$\text{erfc}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2} dt \quad (163)$$

Conversion Formulas

$$Q(x) = \frac{1}{2} \text{erfc}\left(\frac{x}{\sqrt{2}}\right) \quad (164)$$

$$\text{erfc}(x) = 2Q(\sqrt{2}x) \quad (165)$$

Relations with Error Function

$$\text{erfc}(x) = 1 - \text{erf}(x) \quad (166)$$

$$Q(x) = \frac{1}{2} \left[1 - \text{erf} \left(\frac{x}{\sqrt{2}} \right) \right] \quad (167)$$

Approximate equation for $\text{erfc}(x)$ when value of x is large

$$\text{erfc}(x) \approx \frac{e^{-x^2}}{\sqrt{\pi}}, \quad x \gg 1 \quad (168)$$

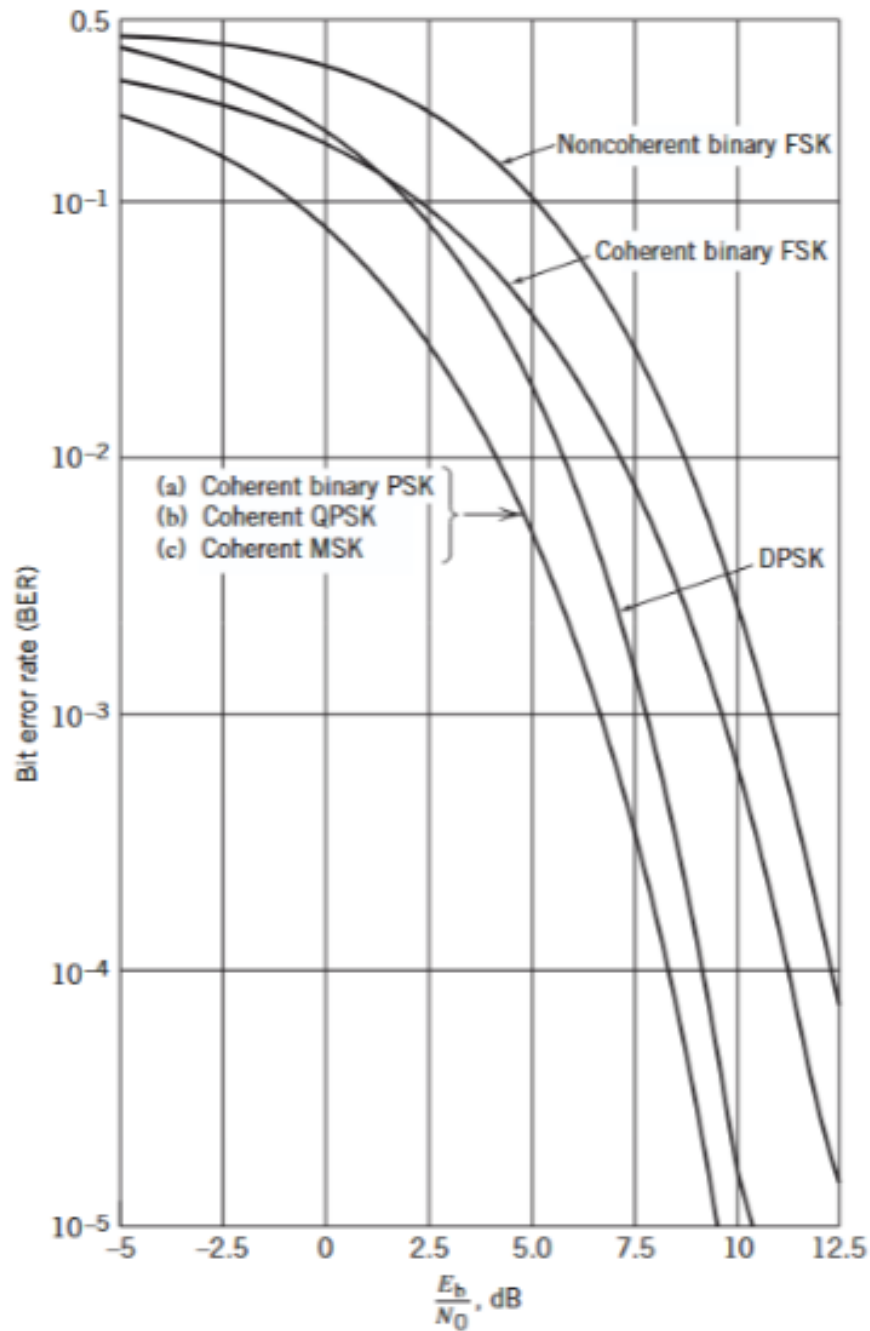


Figure 17: Comparison of the noise performance of different PSK and FSK schemes